Chapter 7
Search and Unemployment

7.1. Introduction

This chapter describes a search problem that is useful for studying unemployment. Search theory puts unemployed workers in a setting where they sometimes choose to reject available offers and to remain unemployed now because they prefer to wait for better offers later. We use the theory to study how workers respond to variations in the rate of unemployment compensation, the perceived riskiness of wage distributions, the probability of being fired, the quality of information about jobs, and the frequency with which a wage distribution can be sampled.

We seek to convey some of the excitement that Robert E. Lucas, Jr. (1987, p.57) expressed when he wrote this about the McCall search model: "Questioning a McCall worker is like having a conversation with an out-of-work friend: 'Maybe you are setting your sights too high' or 'Why did you quit your old job before you had a new one lined up?' This is real social science: an attempt to model, to *understand*, human behavior by visualizing the situations people find themselves in, the options they face and the pros and cons as they themselves see them."

We describe techniques used in the search literature and examples of search models. We study ideas introduced in important papers by McCall (1970) and Jovanovic (1979a). These papers differ in the possibilities with which they confront an unemployed worker.[1] We also study a related model of occupational choice by Neal (1999). The modifications of the basic McCall model by Jovanovic, Neal, and in the various sections and exercises of this chapter all come from visualizing aspects of the situations in which workers find themselves.

[1] Stigler's (1961) early paper studied a setting different from both McCall's and Jovanovic's. In Stigler's model, an unemployed worker has to choose in advance a number n of offers to draw, from which he takes the highest wage offer. Stigler's formulation of the search problem was not sequential.

7.2. McCall's model of intertemporal job search

McCall (1969) studied an unemployed worker who is searching for a job under the following circumstances: Each period the worker draws one offer w from the same wage distribution $F(W) = \mathrm{Prob}\{w \leq W\}$, with $F(0) = 0$, $F(B) = 1$ for $B < \infty$.[2] The worker has the option of rejecting the offer, in which case he or she receives c in unemployment compensation this period and waits until next period to draw another offer from F; alternatively, the worker can accept the offer to work at w, in which case he or she receives a wage of w per period forever. Neither quitting nor firing is permitted. Offers rejected in the past cannot be recalled.

Let y_t be the worker's income in period t. We have assumed that $y_t = c$ if the worker is unemployed and $y_t = w$ if the worker has accepted an offer to work at wage w. The unemployed worker wants to maximize the mathematical expectation of $\sum_{t=0}^{\infty} \beta^t y_t$ where $0 < \beta < 1$ is a discount factor. Note that the worker *chooses* the probability distribution over income sequences $\{y_t\}_{t=0}^{\infty}$ with respect to which the criterion $E \sum_{t=0}^{\infty} \beta^t y_t$ is evaluated.

Let $v(w)$ be the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$ for a previously unemployed worker who has offer w in hand, who is deciding whether to accept or to reject it, and who behaves optimally. The value function $v(w)$ satisfies the Bellman equation

$$v(w) = \max_{\text{accept,reject}} \left\{ \frac{w}{1-\beta}, c + \beta \int_0^B v(w')\, dF(w') \right\}, \tag{7.2.1}$$

where maximization is over values of outcomes associated with two actions: (1) *accept* the wage offer w and work forever at wage w, or (2) *reject* the offer, receive c this period, and draw a new offer w' from distribution F next period. The value of accepting the offer is $\frac{w}{1-\beta}$, the present value of the constant wage. The value of rejecting the offer is the unemployment compensation c received today plus the discounted expected value $\beta \int_0^B v(w') dF(w')$ of drawing a new offer and deciding optimally tomorrow. Figure 7.2.1 graphs the functional

[2] Appendix A of this chapter describes useful mathematical properties of nonnegative random variables.

equation (7.2.1) and reveals that its solution is of the form

$$v(w) = \begin{cases} \dfrac{\overline{w}}{1-\beta} = c + \beta \displaystyle\int_0^B v(w')\, dF(w') & \text{if} \quad w \le \overline{w} \\ \dfrac{w}{1-\beta} & \text{if} \quad w \ge \overline{w}. \end{cases} \tag{7.2.2}$$

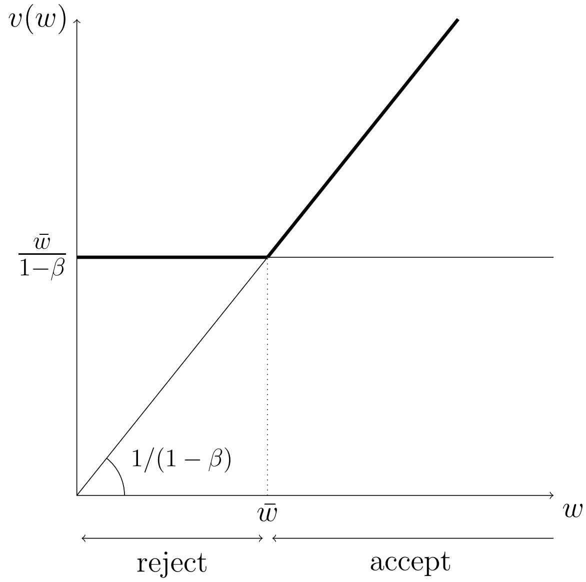

Figure 7.2.1: The function $v(w) = \max\{w/(1-\beta), c + \beta\int_0^B v(w')dF(w')\}$. The reservation wage $\overline{w} = (1-\beta)[c + \beta\int_0^B v(w')dF(w')]$.

Using equation (7.2.2), we can convert the functional equation (7.2.1) in the value function $v(w)$ into an ordinary equation in the reservation wage $\overline{w}$. Evaluating $v(\overline{w})$ and using equation (7.2.2), we have

$$\frac{\overline{w}}{1-\beta} = c + \beta \int_0^{\overline{w}} \frac{\overline{w}}{1-\beta} dF(w') + \beta \int_{\overline{w}}^B \frac{w'}{1-\beta} dF(w')$$

or

$$\begin{aligned}
&\frac{\overline{w}}{1-\beta}\int_0^{\overline{w}} dF\left(w'\right)+\frac{\overline{w}}{1-\beta}\int_{\overline{w}}^{B} dF\left(w'\right) \\
&\qquad = c+\beta\int_0^{\overline{w}}\frac{\overline{w}}{1-\beta}dF\left(w'\right)+\beta\int_{\overline{w}}^{B}\frac{w'}{1-\beta}dF\left(w'\right)
\end{aligned}$$

or

$$\overline{w}\int_0^{\overline{w}} dF\left(w'\right)-c=\frac{1}{1-\beta}\int_{\overline{w}}^{B}\left(\beta w'-\overline{w}\right)dF\left(w'\right).$$

Adding $\overline{w}\int_{\overline{w}}^{B} dF(w')$ to both sides gives

$$\left(\overline{w}-c\right)=\frac{\beta}{1-\beta}\int_{\overline{w}}^{B}\left(w'-\overline{w}\right)dF\left(w'\right). \tag{7.2.3}$$

The left side is the opportunity cost of *rejecting* an offer $\overline{w}$. The right side is the opportunity cost of *accepting* an offer $\overline{w}$, which equals the probability that next period's wage draw w' exceeds $\overline{w}$ times the expected present value of $w'-\overline{w}$ conditional on drawing an offer w' greater than $\overline{w}$. To see this, rearrange and rewrite equation (7.2.3) to become

$$\overline{w}=c+\text{Prob}\left(w\geq\overline{w}\right)\frac{\beta}{1-\beta}\int_{\overline{w}}^{B}\left(w'-\overline{w}\right)d\tilde{F}\left(w'\right). \tag{7.2.4}$$

where $\tilde{F}(w)=\frac{F(w')}{\text{Prob}(w\geq\overline{w})}$ is the cdf of w conditional on $w\geq\overline{w}$. The term after the $+$ sign on the right side of equation (7.2.4) is the value that the worker assigns to an option to wait for an offer better than $\overline{w}$. It equals the probability that $w\geq\overline{w}$ times the discounted present value of a constant continuation income stream that starts next period and that equals the expected value of the conditional wage distribution $\tilde{F}$. Notice that (7.2.4) is one equation in the unknown reservation wage $\overline{w}$, so while it might be useful for interpreting why the worker chooses to reject some offers that exceed $\overline{w}$, it is not a formula for $\overline{w}$ in terms of the underlying constituents c, F, β.

7.2.1. Characterizing reservation wage

Define the function on the right side of equation (7.2.3) as

$$h(w) = \frac{\beta}{1-\beta}\int_w^B (w' - w)\, dF(w'). \tag{7.2.5}$$

Notice that $h(0) = Ew\beta/(1-\beta)$, that $h(B) = 0$, and that $h(w)$ is differentiable, with derivative[3]

$$h'(w) = -\frac{\beta}{1-\beta}[1 - F(w)] < 0$$

and second derivative

$$h''(w) = \frac{\beta}{1-\beta}F'(w) > 0,$$

so that $h(w)$ is convex to the origin. Figure 7.2.2 graphs $h(w)$ against $(w - c)$ and indicates how $\overline{w}$ is determined. From Figure 7.2.2 it is apparent that an increase in unemployment compensation c leads to an increase in $\overline{w}$.

To get another useful characterization of $\overline{w}$, express equation (7.2.3) as

$$\begin{aligned}
\overline{w} - c =& \frac{\beta}{1-\beta}\int_{\overline{w}}^B (w' - \overline{w})\, dF(w') + \frac{\beta}{1-\beta}\int_0^{\overline{w}} (w' - \overline{w})\, dF(w') \\
& - \frac{\beta}{1-\beta}\int_0^{\overline{w}} (w' - \overline{w})\, dF(w') \\
=& \frac{\beta}{1-\beta}Ew - \frac{\beta}{1-\beta}\overline{w} - \frac{\beta}{1-\beta}\int_0^{\overline{w}} (w' - \overline{w})\, dF(w')
\end{aligned}$$

or

$$\overline{w} - (1-\beta)c = \beta Ew - \beta\int_0^{\overline{w}} (w' - \overline{w})\, dF(w').$$

Applying integration by parts to the last integral on the right side and rearranging, we have

$$\overline{w} - c = \beta(Ew - c) + \beta\int_0^{\overline{w}} F(w')\, dw'. \tag{7.2.6}$$

[3] To compute $h'(w)$, apply Leibniz's rule to equation (7.2.5). Let $\phi(t) = \int_{\alpha(t)}^{\beta(t)} f(x,t)dx$ for $t \in [c,d]$. Assume that f and f_t are continuous and that α, β are differentiable on $[c,d]$. Then Leibniz's rule asserts that $\phi(t)$ is differentiable on $[c,d]$ and

$$\phi'(t) = f[\beta(t), t]\beta'(t) - f[\alpha(t), t]\alpha'(t) + \int_{\alpha(t)}^{\beta(t)} f_t(x,t)\, dx.$$

To apply this formula to the equation in the text, let w play the role of t.

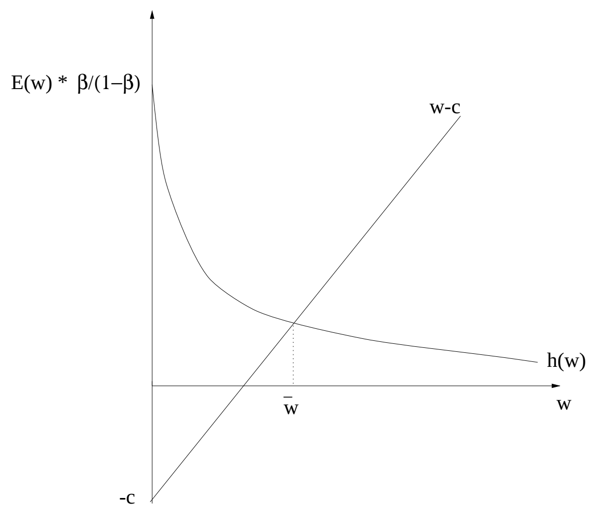

Figure 7.2.2: The reservation wage $\overline{w}$ that satisfies $\overline{w}-c = [\beta/(1-\beta)]\int_{\overline{w}}^{B}(w'-\overline{w})dF(w') \equiv h(\overline{w})$.

At this point it is useful to define the function

$$g\left(s\right)=\int_0^s F\left(p\right)dp. \tag{7.2.7}$$

This function has the characteristics that $g(0)=0$, $g(s)\geq 0$, $g'(s)=F(s)>0$, and $g''(s)=F'(s)>0$ for $s>0$. Then equation (7.2.6) can be represented as $\overline{w}-c=\beta(Ew-c)+\beta g(\overline{w})$. Figure 7.2.3 uses equation (7.2.6) to determine $\overline{w}$.

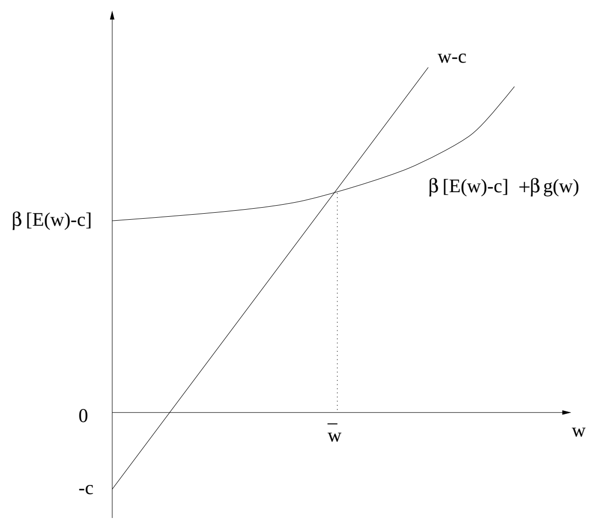

Figure 7.2.3: The reservation wage, $\overline{w}$, that satisfies $\overline{w}-c = \beta(Ew-c)+\beta\int_0^{\overline{w}} F(w')dw' \equiv \beta(Ew-c)+\beta g(\overline{w})$.

7.2.2. Effects of mean-preserving spreads

Here we exploit properties of nonnegative random variables summarized in appendix A of this chapter. Figure 7.2.3 can be used to establish two propositions about $\overline{w}$. First, given F, $\overline{w}$ increases when the rate of unemployment compensation c increases. Second, given c, a mean-preserving increase in risk causes $\overline{w}$ to increase. This second proposition follows directly from Figure 7.2.3 and characterization (iii) or (v) of a mean-preserving increase in risk from appendix A. From the definition of g in equation (7.2.7) and the characterization (iii) or (v), a mean-preserving spread causes an upward shift in $\beta(Ew-c)+\beta g(w)$.

Since an increase in unemployment compensation and a mean-preserving increase in risk both raise the reservation wage, it follows from the expression for the value function in equation (7.2.2) that unemployed workers are also better off with both such increases. It is obvious that an increase in unemployment compensation raises the welfare of unemployed workers but it might seem surprising that a mean-preserving increase in risk does too. Intuition for this latter finding can be gleaned from the result in option pricing theory that the value of an option is an increasing function of the variance in the price of the underlying asset. This is so because the option holder chooses to accept payoffs only from the right tail of the distribution. In our context, the unemployed worker has the option to accept a job and the asset value of a job offering wage rate w is equal to $w/(1-\beta)$. Under a mean-preserving increase in risk, the higher probability attached to very good wage offers increases the value of searching for a job while the higher probability attached to very bad wage offers is not detrimental because the option to work will not be exercised at such low wages.

7.2.3. Allowing quits

Thus far, we have supposed that the worker cannot quit. It happens that had we allowed the worker to quit and search again, after being unemployed one period, he would never exercise that option. To see this point, recall that the reservation wage $\overline{w}$ in (7.2.2) satisfies

$$v(\overline{w}) = \frac{\overline{w}}{1-\beta} = c + \beta \int v(w')\, dF(w'). \tag{7.2.8}$$

Suppose the worker has in hand an offer to work at wage w. Assuming that the worker behaves optimally after rejecting a wage w, we can compute the lifetime utility associated with three mutually exclusive alternative ways of responding to that offer:

A1. Accept the wage and keep the job forever:

$$\frac{w}{1-\beta}.$$

A2. Accept the wage but quit after t periods:

$$\frac{w-\beta^t w}{1-\beta} + \beta^t \left(c + \beta \int v(w')\, dF(w') \right) = \frac{w}{1-\beta} - \beta^t \frac{w-\overline{w}}{1-\beta}.$$

A3. Reject the wage:

$$c + \beta \int v(w')\, dF(w') = \frac{\overline{w}}{1-\beta}.$$

We conclude that if $w < \overline{w}$,

$$A1 \prec A2 \prec A3,$$

and if $w > \overline{w}$,

$$A1 \succ A2 \succ A3.$$

The three alternatives yield the same lifetime utility when $w = \overline{w}$.

7.2.4. Waiting times

It is straightforward to derive the probability distribution of the waiting time until a job offer is accepted. Let N be the random variable "length of time until a successful offer is encountered," with the understanding that $N = 1$ if the first job offer is accepted. Let $\lambda = \int_0^{\overline{w}} dF(w')$ be the probability that a job offer is rejected. Then we have $\text{Prob}\{N = 1\} = (1 - \lambda)$. The event that $N = 2$ is the event that the first draw is less than $\overline{w}$, which occurs with probability λ, and that the second draw is greater than $\overline{w}$, which occurs with probability $(1-\lambda)$. By virtue of the independence of successive draws, we have $\text{Prob}\{N = 2\} = (1 - \lambda)\lambda$. More generally, $\text{Prob}\{N = j\} = (1 - \lambda)\lambda^{j-1}$, so the waiting time is geometrically distributed. The mean waiting time $\bar{N}$ is given by

$$\begin{aligned}\bar{N} &= \sum_{j=1}^{\infty} j \cdot \text{ Prob}\{N = j\} = \sum_{j=1}^{\infty} j\,(1-\lambda)\,\lambda^{j-1} = (1-\lambda)\sum_{j=1}^{\infty}\sum_{k=1}^{j}\lambda^{j-1} \\ &= (1-\lambda)\sum_{k=0}^{\infty}\sum_{j=1}^{\infty}\lambda^{j-1+k} = (1-\lambda)\sum_{k=0}^{\infty}\lambda^{k}\,(1-\lambda)^{-1} = (1-\lambda)^{-1}.\end{aligned}$$

That is, the mean waiting time to a successful job offer equals the reciprocal of the probability of accepting an offer on a single trial.[4]

To illustrate the power of a recursive approach, we can also compute the mean waiting time $\bar{N}$ as follows. First, because the environment is stationary and associated with a constant reservation wage and a constant probability of escaping unemployment, it follows that in any period the "remaining" mean waiting time for all unemployed workers equals $\bar{N}$. That is, all unemployed

[4] An alternative way of deriving the mean waiting time is to use the algebra of z transforms. Define $h(z) = \sum_{j=0}^{\infty} h_j z^j$ and note that $h'(z) = \sum_{j=1}^{\infty} j h_j z^{j-1}$ and $h'(1) = \sum_{j=1}^{\infty} j h_j$. (For an introduction to z transforms, see Gabel and Roberts, 1973.) The z transform of the sequence $(1-\lambda)\lambda^{j-1}$ is given by $\sum_{j=1}^{\infty}(1-\lambda)\lambda^{j-1}z^j = (1-\lambda)z/(1-\lambda z)$. Evaluating $h'(z)$ at $z = 1$ gives, after some simplification, $h'(1) = 1/(1-\lambda)$. Therefore, we have that the mean waiting time is $(1-\lambda)\sum_{j=1}^{\infty} j\lambda^{j-1} = 1/(1-\lambda)$.

workers face a mean waiting time of $\bar{N}$ regardless of how long of an unemployment spell they have endured. Second, the mean waiting time $\bar{N}$ must then be equal to the weighted sum of two possible outcomes: either the worker accepts a job next period, with probability $(1-\lambda)$; or she remains unemployed in the next period, with probability λ. In the first case, the worker will have ended her unemployment after one last period of unemployment while in the second case, the worker will have suffered one period of unemployment *and* will face a remaining mean waiting time of $\bar{N}$ periods. Hence, the mean waiting time must satisfy:

$$\bar{N} = (1-\lambda)\cdot 1 \,+\, \lambda \cdot \left(1+\bar{N}\right) \qquad \Longrightarrow \qquad \bar{N} = (1-\lambda)^{-1}\,.$$

We invite the reader to prove that, given F, the mean waiting time increases with increases in the rate of unemployment compensation, c.

7.2.5. Firing

We now consider a modification of the job search model in which each period after the first period on the job the worker faces probability α of being fired, where $1 > \alpha > 0$. The probability α of being fired next period is assumed to be independent of tenure. A previously unemployed worker samples wage offers from a time-invariant and known probability distribution F. Unemployed workers receive unemployment compensation in the amount c each period. A worker receives a time-invariant wage w on a job until she is fired. Only previously employed workers are fired. A previously employed worker who is fired at the beginning of a period cannot draw a new wage offer that period but must be unemployed for one period.

Let $\hat{v}(w)$ be the expected present value of income of a previously unemployed worker who has offer w in hand and who behaves optimally. Thus, if she rejects the offer, she receives c in unemployment compensation this period and next period draws a new offer w' whose value to her now is $\beta \int \hat{v}(w')dF(w')$. If she rejects the offer, $\hat{v}(w) = c + \beta \int \hat{v}(w')dF(w')$. If she accepts the offer, she receives w this period; next period with probability $1-\alpha$, she is not fired and gets a continuation value $\hat{v}(w)$ next period that this period is worth $\beta\hat{v}(w)$; with probability α, she is fired next period, which has the consequence that after one period of unemployment she draws a new wage, an outcome that today is worth $\beta[c+\beta\int\hat{v}(w')dF(w')]$. Therefore, if she accepts the offer

$$\hat{v}\left(w\right)=w+\beta\left(1-\alpha\right)\hat{v}\left(w\right)+\beta\alpha\left[c+\beta\int\hat{v}\left(w'\right)dF\left(w'\right)\right].$$

A previously unemployed worker's value function $\hat{v}(w)$ after drawing w this period satisfies the Bellman equation[5]

$$\hat{v}\left(w\right)=\max_{\text{accept,reject}}\Big\{w+\beta\left(1-\alpha\right)\hat{v}\left(w\right)+\beta\alpha\left[c+\beta E\hat{v}\right],\ c+\beta E\hat{v}\Big\},$$

where $E\hat{v}=\int\hat{v}(w')dF(w')$. Here the appearance of $\hat{v}(w)$ on the right side recognizes that if the worker had accepted wage offer w last period with expected discounted present value $\hat{v}(w)$, the stationarity of the problem (i.e., the fact that F,α,c are all fixed) makes $\hat{v}(w)$ also be the continuation value associated with retaining this job next period. The Bellman equation has a solution of the form[6]

$$\hat{v}\left(w\right)=\begin{cases}\dfrac{w+\beta\alpha\left[c+\beta E\hat{v}\right]}{1-\beta\left(1-\alpha\right)}, & \text{if } w\geq\overline{w}\\ c+\beta E\hat{v}, & w\leq\overline{w}\end{cases}$$

where $\overline{w}$ solves

$$\frac{\overline{w}+\beta\alpha\left[c+\beta E\hat{v}\right]}{1-\beta\left(1-\alpha\right)}=c+\beta E\hat{v},$$

which can be rearranged as

$$\frac{\overline{w}}{1-\beta}=c+\beta\int\hat{v}\left(w'\right)dF\left(w'\right). \tag{7.2.9}$$

We can compare the reservation wage in (7.2.9) to the reservation wage in expression (7.2.8) that prevailed when there was no risk of being fired. The

[5] If a worker who is fired at the beginning of a period were to have the opportunity to draw a new offer that same period, then the Bellman equation would instead be

$$\tilde{v}\left(w\right)=\max_{\text{accept,reject}}\Big\{w+\beta\left(1-\alpha\right)\tilde{v}\left(w\right)+\beta\alpha\int\tilde{v}\left(w'\right)dF\left(w'\right),c+\beta\int\tilde{v}\left(w'\right)dF\left(w'\right)\Big\}.$$

[6] That it takes this form can be established by guessing that $\hat{v}(w)$ is nondecreasing in w. This guess implies the equation in the text for $\hat{v}(w)$, which is nondecreasing in w. This argument verifies that $\hat{v}(w)$ is nondecreasing, given the uniqueness of the solution of the Bellman equation.

two expressions look identical but the reservation wages differ because the value functions differ. In particular, $\hat{v}(w)$ is strictly less than $v(w)$. This is an immediate implication of our argument that it cannot be optimal to quit if you have accepted a wage strictly greater than the reservation wage in the situation without possible firings (see section 7.2.3). So even though workers who face a zero probability of being fired can mimic outcomes from a setting in which they would face a positive probability of being fired by occasionally "firing themselves" by quitting into unemployment, they choose not to do that because it would lower the expected present value of their income stream. Since employed workers who might be fired with positive probability are worse off than employed workers who can never be fired, it follows that $\hat{v}(w)$ lies strictly below $v(w)$ for all values of w. Even at wages w that are rejected, the value function partly reflects a stream of future outcomes whose mathematical expectation is less favorable when a worker faces a chance of being fired.

Since the value function $\hat{v}(w)$ with firings lies strictly below the value function $v(w)$ without firings, it follows from (7.2.9) and (7.2.8) that the reservation wage $\overline{w}$ is strictly lower when there is a positive probability of being fired. There is less of a reason to wait for high-paying jobs when a worker typically expects a job to last for a shorter period of time. That is, a higher probability of being fired reduces the amount of time that an unemployed worker should optimally invest less time in search.

7.3. A lake model

Consider an economy consisting of a continuum of *ex ante* identical workers living in the environment described in the previous section. These workers move recurrently between unemployment and employment. The mean duration of each spell of employment is α^{-1} and the mean duration of unemployment is $[1 - F(\overline{w})]^{-1}$. The average unemployment rate U_t across the continuum of workers obeys the difference equation

$$U_{t+1} = \alpha\,(1 - U_t) + F\,(\overline{w})\,U_t,$$

where α is the hazard rate of escaping employment and $[1 - F(\overline{w})]$ is the hazard rate of escaping unemployment. Solving the above difference equation for a stationary solution, i.e., imposing $U_{t+1} = U_t = U$, gives

$$U = \frac{\alpha}{\alpha + 1 - F(\overline{w})} \qquad \Longrightarrow \qquad U = \frac{\frac{1}{1-F(\overline{w})}}{\frac{1}{1-F(\overline{w})} + \frac{1}{\alpha}}. \tag{7.3.1}$$

Equation (7.3.1) expresses a stationary unemployment rate in terms of the ratio of the average duration of unemployment to the sum of average durations of unemployment and employment. The unemployment rate, being an average across workers at each moment, thus reflects the average outcomes experienced by workers *across time*. This way of linking economy-wide averages at a point in time with the time-series average for a representative worker is our first encounter with a class of Bewley models that we shall study in chapter 19.

This model of unemployment is sometimes called a lake model and can be depicted as in Figure 7.3.1, with two lakes denoted U and $1-U$ that represent volumes of unemployment and employment, and flows of rate α from the $1-U$ lake to the U lake and of rate $1-F(\overline{w})$ from the U lake to the $1-U$ lake. Equation (7.3.1) allows us to study determinants of the unemployment rate in terms of the hazard rate α of becoming unemployed and the hazard rate $1-F(\overline{w})$ of escaping unemployment.

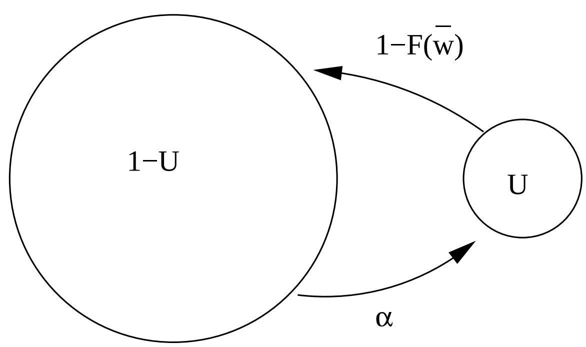

Figure 7.3.1: Lake model with flows of rate α from employment state $1-U$ to unemployment state U and of rate $[1-F(\overline{w})]$ from U to $1-U$.

7.4. Offer distribution unknown

Consider the following modification of the McCall search model. An unemployed worker wants to maximize the expected present value of $\sum_{t=0}^{\infty} \beta^t y_t$ where y_t equals wage w when employed and unemployment compensation c when unemployed. Each period the worker receives one offer to work forever at a wage w drawn from one of two possible cumulative distribution functions F or G, where $F(0) = G(0) = 0$ and $F(B) = G(B) = 1$ for $B > 0$. We assume that the distributions have densities f and g, respectively, and that they have common support. Nature draws from the same distribution, either F or G, at all dates. The worker knows this but does not know whether it is F or G. At time 0 *before* drawing a wage offer, the worker attaches probability $\pi_{-1} \in (0,1)$ to the distribution being F. Before drawing a wage at time 0, the worker thus believes that the density of w_0 is $h(w_0;\pi_{-1}) = \pi_{-1} f(w_0) + (1-\pi_{-1}) g(w_0)$. Let $a \in \{f, g\}$ be an index that indicates whether nature chose permanently to draw from distribution f or from distribution g. After drawing w_0, the worker uses Bayes' law to deduce that the posterior probability $\pi_0 = \text{Prob}(a = f)|w_0$ that the density is $f(w)$ is[7]

$$\pi_0 = \frac{\pi_{-1} f(w_0)}{\pi_{-1} f(w_0) + (1-\pi_{-1}) g(w_0)}.$$

More generally, after making the tth draw and having observed $w_t, w_{t-1}, \ldots, w_0$, the worker believes that the probability that w_{t+1} will be drawn from distribution F is

$$\pi_t = \pi_t(w_t|\pi_{t-1}) \equiv \frac{\pi_{t-1} f(w_t)/g(w_t)}{\pi_{t-1} f(w_t)/g(w_t) + (1-\pi_{t-1})} \tag{7.4.1}$$

and that the density of w_{t+1} conditional on $w_t, w_{t-1}, \ldots, w_0$ is

$$h(w_{t+1};\pi_t) = \pi_t f(w_{t+1}) + (1-\pi_t) g(w_{t+1}). \tag{7.4.2}$$

[7] The worker's initial beliefs induce a joint probability distribution over a potentially infinite sequence of draws $w_0, w_1, \ldots$. Bayes' law is simply an application of the laws of probability to compute the conditional distribution of the tth draw w_t conditional on $[w_0, \ldots, w_{t-1}]$. Since we assume from the start that the decision maker *knows* the joint distribution and the laws of probability, one respectable view is that Bayes' law is less a 'theory of learning' than a statement about the consequences of information inflows for a decision maker who thinks he knows the truth (i.e., a joint probability distribution) from the beginning.

Notice that

$$\begin{aligned} E(\pi_t|\pi_{t-1}) &= \int \Big[\frac{\pi_{t-1} f(w)}{\pi_{t-1} f(w) + (1-\pi_{t-1}) g(w)}\Big] \Big[\pi_{t-1} f(w) + (1-\pi_{t-1}) g(w)\Big] dw \\ &= \pi_{t-1} \int f(w) dw \\ &= \pi_{t-1}, \end{aligned}$$

so that the process π_t is a *martingale* bounded by 0 and 1. (In the first line in the above string of equalities, the term in the first set of brackets is just π_t as a function of w_t, while the term in the second set of brackets is the density of w_t conditional on $w_{t-1}, \ldots, w_0$ or equivalently conditional on the statistic π_{t-1} that completely summarizes information that history $w_{t-1}, \ldots, w_0$ contains about whether nature takes i.i.d. draws from cdf F or cdf G.[8] Here we are computing $E(\pi_t|\pi_{t-1})$ under the subjective density described in the second term in brackets.[9] It is also possible to show that under the f distribution

$$E(\pi_t|\pi_{t-1}) = \int \left[\frac{\pi_{t-1} f(w)}{\pi_{t-1} f(w) + (1-\pi_{t-1}) g(w)}\right] f(w)\, dw \geq \pi_{t-1}$$

and that under the g distribution

$$E(\pi_t|\pi_{t-1}) = \int \left[\frac{\pi_{t-1} f(w)}{\pi_{t-1} f(w) + (1-\pi_{t-1}) g(w)}\right] g(w)\, dw \leq \pi_{t-1}.$$

These two inequalities can be shown to imply that when nature permanently draws from f, $\pi_t \to 1$, while when nature permanently draws from g, $\pi_t \to 0$.

Let $v(w_t, \pi_t)$ be the optimal value of the problem for a previously unemployed worker who has just drawn w and who updates π according to (7.4.1). The optimal value function $v(w, \pi_t)$ satisfies the Bellman equation

$$v(w, \pi_t) = \max_{\text{accept,reject}} \left\{ \frac{w}{1-\beta}, c + \beta \int v(w', \pi_{t+1}(w'))\, h(w'; \pi_t)\, dw' \right\} \tag{7.4.3}$$

[8] Thus, π_{t-1} is a *sufficient statistic* for a parameter $s_{-1} \in \{0,1\}$, where $s_{-1} = 1$ if nature always draws from cdf F and $s_{-1} =$ if nature always draws from cdf G. We can regard s_{-1} as a time-invariant hidden Markov state.

[9] It follows from the martingale convergence theorem (see appendix A of chapter 18) that π_t converges almost surely to a random variable π_∞ in $[0,1]$. Practically, this means that probability one is attached to sample paths $\{\pi_t\}_{t=0}^\infty$ that converge pointwise. Different sample paths can converge to different limiting values, so that limit points of $\{\pi_t\}_{t=0}^\infty$ as $t \to +\infty$ are realizations of a random variable that has a non-trivial distribution. By studying equation (7.4.1), it can be verified that the only possible limit points π_∞ are 1 and 0. The martingale property $E\pi_{t+1}|\pi_t = \pi_t$ then implies that the probability attached to sample paths in which $\pi_\infty = 1$ equals π_{-1}.

where maximization is subject to (7.4.1) and (7.4.2). The state vector is the worker's current draw w and his post-draw estimate of the probability that the distribution is f. The second term on the right side of (7.4.3) integrates the value function evaluated at next period's state vector with respect to the conditional probability distribution $h(w'; \pi_t)$ of next period's draw w'. The value function for next period recognizes that π_{t+1} will be updated in a way that depends on w' via Bayes' law as captured by equation (7.4.1). Evidently, the optimal policy is to set a reservation wage $\bar{w}(\pi_t)$ that depends on π_t.

As an example, we have computed the optimal policy by backward induction assuming that f is a uniform distribution on $[0, 2]$ while g is a beta distribution with parameters $(\alpha, \gamma) = (3, 1.2)$.[10] We set unemployment compensation $c = .6$ and the discount factor $\beta = .95$.[11] The two densities are plotted in figure 7.4.1, which shows that the g density provides better prospects for the worker than does the uniform f density. It stands to reason that the worker's reservation wage falls as the posterior probability π that he places on density f rises, as figure 7.4.2 confirms.

Figure 7.4.3 shows empirical cumulative distribution functions for durations of unemployment and π at time of job acceptance under two alternative assumptions about whether the uniform distribution F or the beta distribution G permanently governs the wage. We constructed these by simulating the model 10,000 times at the parameter values just given, starting from a common initial condition for beliefs $\pi_{-1} = .5$ and assuming that, unbeknownst to the worker, either the uniform density $f(w)$ or the beta density $g(w)$ truly governs successive wage draws. Only when π_t approaches 1 will workers have learned whether nature is drawing from f and not g. Evidently, most workers accept jobs long before a law of large numbers has enough time to teach them for sure which of the two densities from which nature draws wage offers. Thus, workers usually choose not to collect enough observations for them to learn for sure which distribution governs wage offers. In both panels, the lower line shows

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10 The beta distribution for w is characterized by a density $g(w; \alpha, \gamma) \propto w^{\alpha-1}(1-w)^{(\gamma-1)}$, where the factor of proportionality is chosen to make the density integrate to 1.

11 The Matlab programs `search_learn_francisco_3.m` and `search_learn_beta_2.m` perform these calculations.

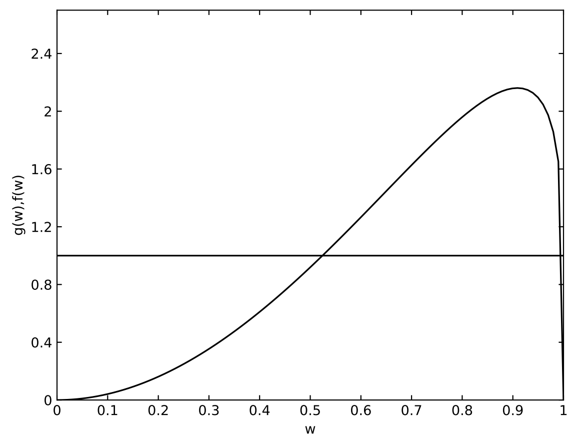

Figure 7.4.1: Two densities for wages, a uniform $f(w)$ and $g(w)$ that is a beta distribution with parameters $(\alpha, \gamma) = (3, 1.2)$.

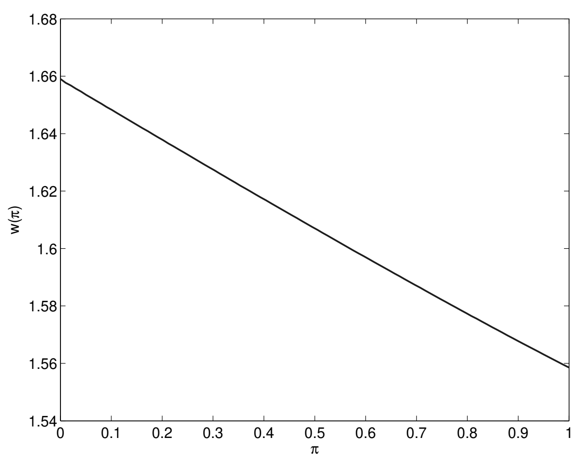

Figure 7.4.2: The reservation wage as a function of the posterior probability π that the worker thinks that the wage is drawn from the uniform density f.

the cumulative distribution function when nature draws from F and the lower panel shows the c.d.f. when nature draws from G.[12]

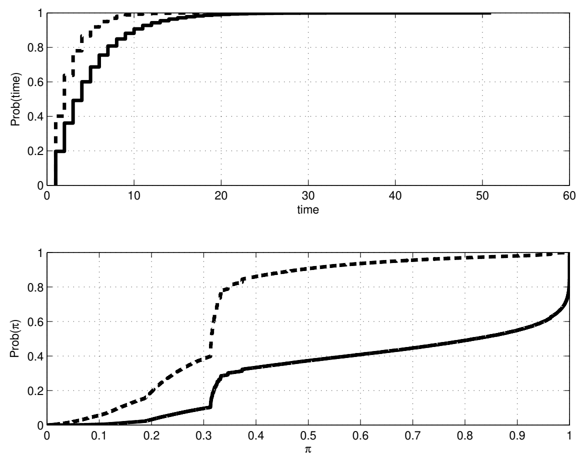

Figure 7.4.3: Top panel: CDF of duration of unemployment; bottom panel: CDF of π at time worker accepts wage and leaves unemployment. In each panel, the lower solid line is the CDF when nature permanently draws from the uniform density f while the dotted line is the CDF when nature permanently draws from the beta density g.

A comparison of the CDF's when nature draws from F and G, respectively, is revealing. When G prevails, the cumulative distribution functions in the top panel reveal that workers typically accept jobs earlier than when F prevails. This captures what the interrogator of an unemployed McCall worker in the passage of Lucas cited in the introduction to this chapter might have had in mind when he said 'Maybe you are setting your sights too high'. The bottom panel reveals that when nature permanently draws from G, employed workers put a higher probability on their having actually sampled from G than from F, while the reverse is true when nature draws permanently from F.

[12] It is a useful exercise to use recall formula $(7.A.2)$ for the mean of a nonnegative random variable and then glance at the CDFs in the bottom panel to approximate the mean π_t at time of job acceptance.

7.5. An equilibrium price distribution

The McCall search model confronts a worker with a given distribution of wages. In this section, we ask why firms might conceivably choose to confront an *ex ante* homogenous collection of workers with a nontrivial distribution of wages. Knowing that the workers have a reservation wage policy, why would a firm ever offer a worker *more* than the reservation wage? That question challenges us to think about whether it is possible to conceive of a coherent setting in which it would be optimal for a collection of profit maximizing firms somehow to make decisions that generate a distribution of wages.

In this section, we take up this question, but for historical reasons investigate it in the context of a sequential search model in which buyers seek the lowest price.[13] Buyers can draw additional offers from a known distribution at a fixed cost c for each additional *batch* of n independent draws from a known price distribution. Both within and across batches, successive draws are independent. The buyer's optimal strategy is to set a reservation price and to continue drawing until the first time a price less than the reservation price has been offered. Let $\tilde{p}$ be the reservation price.

Rothschild (1973) posed the following challenge for a model in which there is a large number of identical buyers each of whom has reservation price $\tilde{p}$. If all sellers know the reservation price $\tilde{p}$, why would any of them offer a price less than $\tilde{p}$? This cogent question points to a force for the price distribution to collapse, an outcome that would destroy the motive for search behavior on the part of buyers. Thus, the challenge is to construct an *equilibrium* version of a search model in which it is in firms' interest to generate the non-trivial price distribution that sustains buyers' search activities.

Burdett and Judd (1983) met this challenge by creating an environment in which *ex ante* identical buyers *ex post* receive differing numbers of price offers that are drawn from a common distribution set by firms. They construct an equilibrium in which a continuum of profit maximizing sellers are content to generate this distribution of prices. Sellers set their prices to maximize expected profit per customer. But sellers don't know the number of other offers that a prospective customer has received. Heterogeneity in the number of offers received by buyers together with seller's ignorance of the number and nature of

[13] See Burdett and Mortensen (1998) for a parallel analysis of the analogous issues in a model of job search.

other offers received by a *particular* customer creates a tradeoff between profit per customer and volume that makes possible a non-degenerate equilibrium price distribution. Firms that post higher prices are lower-volume sellers. Firms that post lower prices are higher-volume sellers. There exists an equilibrium distribution of prices in which all types of firms expect to earn the same profit per potential customer.

7.5.1. A Burdett-Judd setup

A continuum of buyers purchases a single good from one among a continuum of firms. Each firm contacts a fixed measure ν of potential buyers. The firms produce a homogeneous good at zero marginal cost. Each firm takes the c.d.f. of prices charged by other firms as given and chooses a price. The firm wants to maximize its expected profits per consumer. A firm's expected profit per consumer equals its price times the probability that its price is the minimum among the set of acceptable offers received by the buyer. The distribution of prices set by other firms impinges on a firm's expected profits because it affects the probability that its offer will be accepted by a buyer.

7.5.2. Consumer problem with noisy search

A consumer wants to purchase a good for a minimum price. Firms make offers that buyers can view as being drawn from a distribution of nonnegative prices with cumulative distribution function $G(P) = \text{Prob}(p \leq P)$ with $G(\underline{p}) = 0, G(B) = 1$. Assume that G is continuously differentiable and so has an associated probability density. A buyer's search activity is divided into batches. Within each batch the buyer receives a random number of offers drawn from the same distribution G. Burdett and Judd call this structure 'noisy search'. In particular, at a cost of $c > 0$ per search round, with probability $q \in (0,1)$ a buyer receives one offer drawn from G and with probability $1-q$ receives two offers. Thus, a 'round' consists of a 'compound lottery' first of a random number of draws, then that number of i.i.d. random draws price offers from the c.d.f. G. A buyer can recall offers within a round but not across rounds. Evidently, $\text{Prob}\{\min(p_1,p_2) \geq p\} = (1-G(p))^2$ and $\text{Prob}\{\min(p_1,p_2) \leq p\} = 1-(1-G(p))^2$. Then *ex ante* the c.d.f. of low prices drawn in a single round is

$$H(p) = qG(p) + (1-q)\left(1 - (1 - G(p))^2\right). \tag{7.5.1}$$

Let $v(p)$ be the expected price including future search costs of a consumer who has already paid c, has offer p in hand, and is about to decide whether to accept or reject the offer. Let $\tilde{p}$ be the buyer's reservation price. In equilibrium, it will turn out that no seller will over a price higher than $\tilde{p}$. The buyer's Bellman equation is then

$$v(p) = \min_{\text{accept,reject}} \left\{ p, c + \int_{\underline{p}}^{\tilde{p}} v(p')\, dH(p') \right\}. \tag{7.5.2}$$

The reservation price $\tilde{p}$ satisfies $v(\tilde{p}) = \tilde{p} = c + \int_{\underline{p}}^{\tilde{p}} p' dH(p')$, which implies[14]

$$c = \int_{\underline{p}}^{\tilde{p}} H(p)\, dp. \tag{7.5.3}$$

Next note that $E(p) = \int_{\underline{p}}^{\tilde{p}} p dH(p) = pH(p)|_{\underline{p}}^{\tilde{p}} - \int_{\underline{p}}^{\tilde{p}} H(p)dp = \tilde{p} - \int_{\underline{p}}^{\tilde{p}} H(p)dp$. Combining $E(p) = \tilde{p} - \int_{\underline{p}}^{\tilde{p}} H(p)dp$ with equation (7.5.3) implies that

$$\tilde{p} = c + Ep,$$

which states that the reservation price equals the cost of one additional round of search plus the mean price drawn from one more round of noisy search. The challenge is to construct an equilibrium price distribution G, and thus an implied distribution H, in which most firms choose to post prices less than the buyer's reservation price $\tilde{p}$.

---

14 The Bellman equation implies $\tilde{p} = c + \int_{\tilde{p}}^{B} v(p')dH(p)$, which can be rearranged to become $\int_{\underline{p}}^{\tilde{p}} (\tilde{p} - p)dH(p) = c$. Let $u = \tilde{p} - p$ and $dv = dH(p)$ and apply the integration by parts formula $\int udv = uv - \int vdu$ to the previous equality to get $\int_{\underline{p}}^{\tilde{p}} H(p)dp = c$.

7.5.3. Firms

For simplicity and to focus our attention entirely on the search problem, we assume that the good costs firms nothing to produce. In setting its price, we assume that a firm seeks to maximize expected profit per customer. A firm makes an offer to a customer without knowing whether this is the only offer available to the customer or whether the customer, having drawn two offers, possibly has a lower offer in hand. The firm begins by computing the fraction of its customers who will have received one offer and the fraction of its customers who will have received only one offer. Let there be a large number ν of total potential buyers per batch, consisting of νq persons each of whom receives one offer and $\nu(1-q)$ people each of whom receives two offers. The total number of offers is evidently $\nu(1q+2(1-q)) = \nu(2-q)$. Evidently, the fraction of all offers that is received by customers who have received one offer is $\frac{\nu q}{\nu(2-q)} = \frac{q}{2-q}$. This calculation induces a typical firm to believe that the fraction of its customers who receive one offer is

$$\hat{q} = \frac{q}{2-q} \tag{7.5.4}$$

and the fraction who receive two offers is $1-\hat{q} = \frac{2(1-q)}{2-q}$. The firm regards $\hat{q}$ as its estimate of the probability that a given customer has received only its offer, while it thinks that a fraction $1-\hat{q}$ of its customers has also received a competing offer from another firm.

There is a continuum of firms each of which takes as given a price offer distribution of other firms with c.d.f. $G(p)$, where $G(\underline{p}) = 0, G(\tilde{p}) = 1$. We have assumed that G is differentiable.[15] This distribution satisfies the outcome that in equilibrium no firm makes an offer exceeding the buyer's reservation price $\tilde{p}$. Let $Q(p)$ be the probability that a consumer will accept an offer p, where $\underline{p} \leq p \leq \tilde{p}$. Evidently, a consumer who receives *one* offer $p < \tilde{p}$ will accept it with probability 1. But only a fraction $1-G(p)$ of consumer who receive *two* offers will accept an offer $p < \tilde{p}$. Why? Because $1-G(p)$ is the

[15] Burdett and Judd (1983, p. 959, lemma 1) show that an equilibrium G is differentiable when $q \in (0,1)$ and $\tilde{p} > 0$. Their argument goes as follows. Suppose to the contrary that there is a positive probability attached to a single price $p' \in (0,\tilde{p})$. Consider a firm that contemplates charging p'. When $q < 1$, the firm knows that there is a positive probability that a prospective consumer has received another offer also of p'. If the firm lowers its offer infinitesimally, it can expect to steal that customer and thereby increase its expected profits. Therefore, a decision to charge p' can't maximize expected profits for a typical firm. We have been led to a contradiction by assuming that G has a discontinuity at p'.

fraction of consumers whose *other* offer exceeds p; so a fraction $G(p)$ of two-offer customers who receive offer p will reject it because they have received an offer lower than p. Therefore, the overall probability that a randomly encountered consumer will accept an offer $p \in [\underline{p}, \tilde{p}]$ is

$$Q(p) = \hat{q} + (1 - \hat{q})(1 - G(p)). \tag{7.5.5}$$

7.5.4. Equilibrium

The objects in play are a reservation price $\tilde{p}$ and a value function $v(p)$ for a typical buyer; and a c.d.f. $G(p)$ of prices that is the outcome of the independent price-setting decisions of individual firms and that is taken as given by all buyers and sellers.

DEFINITION: An equilibrium is a c.d.f. of price offers $G(p)$ on domain $[\underline{p}, \tilde{p}]$, a c.d.f. of per-batch price offers to consumers $H(p)$, and a reservation price $\tilde{p}$ such that (i) the c.d.f. of offers to buyers $H(p)$ satisfies (7.5.1); (ii) $\tilde{p}$ is an optimal reservation price for buyers that satisfies $c = \int_{\underline{p}}^{\tilde{p}} dH(p)$; and (iii) firms are indifferent with respect to charging any $p \in [\underline{p}, \tilde{p}]$; therefore, firms choose p by randomizing using $G(p)$.

We confirm an equilibrium by using a guess-and-verify method. Make the following guess for an equilibrium c.d.f. $G(p)$.[16] First, set

$$\underline{p} = \hat{q}\tilde{p} \tag{7.5.6}$$

and then set

$$G(p) = \begin{cases} 0 & \text{if } p \leq \underline{p} \\ 1 - \frac{\tilde{p}-p}{p}\frac{\hat{q}}{1-\hat{q}} & \text{if } p \in [\underline{p}, \tilde{p}] \\ 1 & \text{if } p > \tilde{p}\ . \end{cases} \tag{7.5.7}$$

Under this guess, $Q(p)$ becomes

$$Q(p) = \frac{\tilde{p}\hat{q}}{p} \quad \forall p \in [\underline{p}, \tilde{p}]\ .$$

[16] We can make sure that the buyer's search problem is consistent with this guess by setting c to confirm (7.5.3).

Therefore, the expected profit per customer for a firm that sets price $p \in [\underline{p}, \tilde{p}]$ is

$$pQ\left(p\right) = \tilde{p}\hat{q}, \tag{7.5.8}$$

which is evidently independent of the firm's choice of offer p in the interval $[\underline{p}, \tilde{p}]$. The firm is indifferent about the price it offers on this interval. In particular, notice that the right side of equality (7.5.8) is the product of the fraction of a firm's buyers receiving one offer, $\hat{q}$, times the reservation price $\tilde{p}$. This is the expected profit per customer of a firm that charges the reservation price. The left side of equality (7.5.8) is the product of the price p times probability $Q(p)$ that a buyer will accept price p, which as we have noted equals the expected profit per customer for a firm that sets price p.

We assume that firms randomize over choices of p in such a way that $G(p)$ given by (7.5.7) emerges as the c.d.f. for prices.

7.5.5. Special cases

The Burdett-Judd model isolates forces for the price distribution to collapse and countervailing forces that can sustain a nontrivial price distribution.

1. Consider the special case in which $q = 1$ (and therefore $\hat{q} = 1$). Here, $\underline{p} = \tilde{p}$. Formula (7.5.7) shows that the distribution of prices collapses. This case exhibits the Rothschild challenge with which we began.
2. Next, consider the opposite special case in which $q = 0$ and therefore $\hat{q} = 0$. Here, $\underline{p} = 0$ and the c.d.f. $G(p) = 1 \forall p \in [\underline{p}, \tilde{p}]$. Bertrand competition drives all prices down to the marginal cost of production, which we have assumed to be zero. This case exhibits another force for the price distribution to collapse, again in the spirit of Rothschild's challenge.

3 Finally, consider the general case in which $q \in (0,1)$ and therefore $\hat{q} \in (0,1)$). When q is strictly in the interior of $[\underline{p}, \tilde{p}]$, we can sustain a nontrivial distribution of prices. Firms are indifferent between being high volume, low price sellers and high price, low volume sellers. The equilibrium price distribution $G(p)$ renders a firm's expected profits per prospective customer $pQ(p)$ independent of p.

7.6. Neal's model of career choice

This section describes a model of occupational choice that Derek Neal (1999) used to understand employment histories of recent high school graduates. Neal wanted to explain why young men often switch jobs *and* careers early in their work histories, then later focus their searches for jobs within a single career, and finally settle down in a particular job. Neal's model can be regarded as a simplified version of Brian McCall's (1991) model.

A worker chooses career-job (θ, ϵ) pairs subject to the following conditions: There is no unemployment. The worker's earnings at time t equal $\theta_t + \epsilon_t$, where θ_t is a component specific to a *career* and ϵ_t is a component specific to a particular *job*. The worker maximizes $E\sum_{t=0}^{\infty}\beta^t(\theta_t+\epsilon_t)$. A *career* is a draw of θ from c.d.f. F; a *job* is a draw of ϵ from c.d.f. G. Successive draws are independent, and $G(0)=F(0)=0$, $G(B_\epsilon)=F(B_\theta)=1$. The worker can draw a new career only if he also draws a new job. However, the worker is free to retain his existing career θ, and to draw a new job ϵ'. The worker decides at the beginning of a period whether to stay in a career-job pair inherited from the past, stay in an inherited career but draw a new job, or draw a new career-job pair. There is no opportunity to recall past jobs or careers.

Let $v(\theta,\epsilon)$ be the optimal value of the problem at the beginning of a period for a worker currently having inherited career-job pair (θ,ϵ) and who is about to decide whether to draw a new career and or job. The value function $v(\theta,\epsilon)$ satisfies the Bellman equation

$$
\begin{aligned}
v(\theta,\epsilon) = \max\Big\{&\theta+\epsilon+\beta v(\theta,\epsilon),\ \theta+\int[\epsilon'+\beta v(\theta,\epsilon')]\,dG(\epsilon'),\\
&\int\int[\theta'+\epsilon'+\beta v(\theta',\epsilon')]\,dF(\theta')\,dG(\epsilon')\Big\}. \qquad (7.6.1)
\end{aligned}
$$

Maximization is over three possible actions: (1) retain the present job-career pair; (2) retain the present career but draw a new job; and (3) draw both a new job and a new career. We might nickname these three alternatives 'stay put', 'new job', 'new life'. The value function is increasing in both θ and ϵ.

Figures 7.6.1 and 7.6.2 display the optimal value function and the optimal decision rule for Neal's model where F and G are each distributed according to discrete uniform distributions on $[0,5]$ with 50 evenly distributed discrete values for each of θ and ϵ and $\beta=.95$. We computed the value function by iterating to convergence on the Bellman equation. The optimal policy is characterized by three regions in the (θ, ϵ) space. For high enough values of $\epsilon + \theta$, the worker stays put. For high θ but low ϵ, the worker retains his career but searches for a better job. For low values of $\theta + \epsilon$, the worker finds a new career and a new job. In figures 7.6.1 and 7.6.2, the decision to *retain* both job and career occurs in the high θ, high ϵ region of the state space; the decision to retain career θ but search for a new job ϵ' occurs in the high θ and low ϵ region of the state space; and the decision to 'get a new life' by drawing both a new θ' and a new ϵ' occurs in the low θ, low ϵ region.[17]

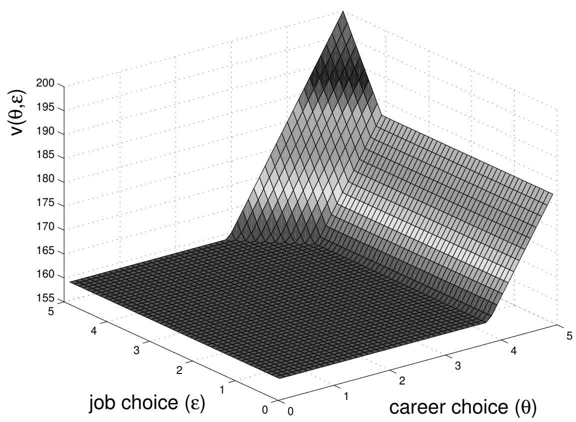

Figure 7.6.1: Optimal value function for Neal's model with $\beta = .95$. The value function is flat in the reject (θ, ϵ) region; increasing in only θ in the keep-career-but-draw-new-job region; and increasing in both θ and ϵ in the stay-put region.

When the career-job pair (θ, ϵ) is such that the worker chooses to stay put, the value function in (7.6.1) attains the value $(\theta + \epsilon)/(1 - \beta)$. Of course, this happens when the decision to stay put weakly dominates the other two actions, which occurs when

$$\frac{\theta + \epsilon}{1 - \beta} \geq \max \{C(\theta), Q\}, \tag{7.6.2}$$

[17] The computations were performed by the Matlab program `neal2.m`.

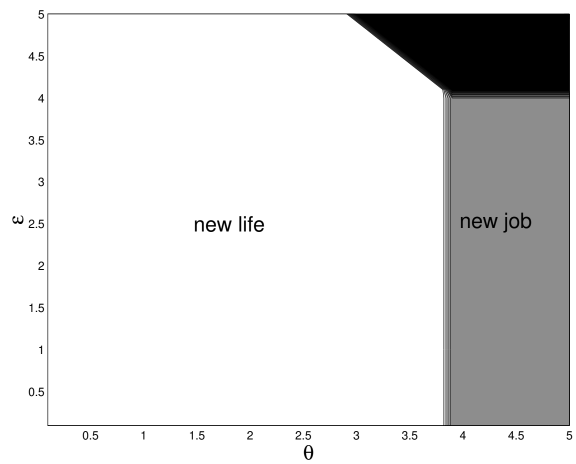

Figure 7.6.2: Optimal decision rule for Neal's model. For (θ, ϵ)'s within the white area, the worker changes both jobs and careers. In the grey region, the worker retains his career but draws a new job. The worker accepts (θ, ϵ) in the black region.

where Q is the value of drawing both a new job and a new career,

$$Q \equiv \int \int \left[\theta' + \epsilon' + \beta v\left(\theta', \epsilon'\right)\right] d\, F\left(\theta'\right) d\, G\left(\epsilon'\right),$$

and $C(\theta)$ is the value of keeping θ but drawing a new job ϵ':

$$C\left(\theta\right) = \theta + \int \left[\epsilon' + \beta v\left(\theta, \epsilon'\right)\right] d\, G\left(\epsilon'\right).$$

For a given career θ, a job $\overline{\epsilon}(\theta)$ makes equation (7.6.2) hold with equality. Evidently, $\overline{\epsilon}(\theta)$ solves

$$\overline{\epsilon}\left(\theta\right) = \max\left[\left(1-\beta\right) C\left(\theta\right) - \theta, \left(1-\beta\right) Q - \theta\right].$$

The decision to stay put is optimal for any career-job pair (θ, ϵ) that satisfies $\epsilon \geq \overline{\epsilon}(\theta)$. When this condition is not satisfied, the worker will draw either a new career-job pair (θ', ϵ') or only a new job ϵ'. Retaining a career θ is optimal when

$$C\left(\theta\right) \geq Q. \tag{7.6.3}$$

We can solve (7.6.3) for the critical career value $\overline{\theta}$ satisfying

$$C\left(\overline{\theta}\right) = Q. \tag{7.6.4}$$

Thus, independently of ϵ, the worker will never abandon any career $\theta \geq \overline{\theta}$. The decision rule for accepting the current career can thus be expressed as follows: accept the current career θ if $\theta \geq \overline{\theta}$ or if the current career-job pair (θ, ϵ) satisfies $\epsilon \geq \overline{\epsilon}(\theta)$.

We can say more about the cutoff value $\overline{\epsilon}(\theta)$ in the retain-θ region $\theta \geq \overline{\theta}$. When $\theta \geq \overline{\theta}$, because we know that the worker will keep θ forever, it follows that

$$C\left(\theta\right) = \frac{\theta}{1-\beta} + \int J\left(\epsilon'\right) dG\left(\epsilon'\right),$$

where $J(\epsilon)$ is the optimal value of $\sum_{t=0}^{\infty} \beta^t \epsilon_t$ for a worker who has already decided to keep career θ, who has just drawn ϵ, and who can draw a new job ϵ' next period. The Bellman equation for J is

$$J\left(\epsilon\right) = \max\left\{\frac{\epsilon}{1-\beta}, \epsilon + \beta \int J\left(\epsilon'\right) dG\left(\epsilon'\right)\right\}. \tag{7.6.5}$$

This resembles the Bellman equation for the optimal value function for the basic McCall model, with a slight modification. The optimal policy is of the reservation-job form: keep the job ϵ if $\epsilon \geq \overline{\epsilon}$, otherwise try a new job next period. The absence of θ from (7.6.5) implies that in the range $\theta \geq \overline{\theta}$, $\overline{\epsilon}$ is independent of θ.

These results explain some features of the value function plotted in Figure 7.6.1 At the boundary separating the "new life" and "new job" regions of the (θ, ϵ) plane, equation (7.6.4) is satisfied. At the boundary separating the "new job" and "stay put" regions, $\frac{\theta+\epsilon}{1-\beta} = C(\theta) = \frac{\theta}{1-\beta} + \int J(\epsilon')dG(\epsilon')$. Finally, between the "new life" and "stay put" regions, $\frac{\theta+\epsilon}{1-\beta} = Q$, which defines a diagonal line in the (θ, ϵ) plane (see Figure 7.6.2). The value function equals the constant Q in the "get a new life" region (i.e., the region in which the optimal decision is to draw a new (θ, ϵ) pair). Equation (7.6.3) helps us understand why there is a set of high θ's in Figure 7.6.2 for which $v(\theta, \epsilon)$ rises with θ but is flat with respect to ϵ.

An interesting feature of the model is that it is possible to draw a (θ, ϵ) pair that makes the value of keeping the career (θ) and drawing a new job match (ϵ') exceed both the value of stopping search and the value of starting again to search from the beginning by drawing a new (θ', ϵ') pair. This outcome occurs when a large θ is drawn with a small ϵ. In this case, it can occur that $\theta \geq \overline{\theta}$ and $\epsilon < \overline{\epsilon}(\theta)$.[18]

Viewed as a normative model for young workers, Neal's model tells them: don't shop for a firm until you have found a career you like. As a positive model, it predicts that workers will not switch careers after they have settled on one. Neal presents data indicating that while this stark prediction does not hold up perfectly, it is a good first approximation. He suggests that extending the model to include learning, along the lines of Jovanovic's model to be described in section 7.7, could help explain the later career switches that his model misses.[19]

7.7. Jovanovic's matching model

Another interesting response to Rothschild's questions about the source of the equilibrium wage (or price) distribution comes from matching models. Here the main idea is to reinterpret w not as a wage but instead, more broadly, as a parameter characterizing the entire quality of a match occurring between a pair of agents. The variable w is regarded as a summary measure of the productivities and utilities jointly generated by the activities of the match. We can consider pairs consisting of a firm and a worker, a man and a woman, a house and an owner, or a person and a hobby. The idea is to analyze the way in which matches form and dissolve by viewing both parties to a match as being drawn from populations that are statistically homogeneous to an outside observer, even though the match is idiosyncratic from the perspective of the parties to the match.

Jovanovic (1979a) used a model of this kind supplemented by an hypothesis that both sides of a match behave optimally but only gradually learn about the quality of the match. Jovanovic was motivated by a desire to explain three features of labor market data: (1) on average, wages rise with tenure on the job, (2) quits are negatively correlated with tenure (that is, a quit has a higher

[18] Pavan (2011) builds on Neal's model in interesting ways.

[19] Neal's model can be used to deduce waiting times to the event $(\theta \geq \overline{\theta}) \cup (\epsilon \geq \overline{\epsilon}(\theta))$. The first event within the union is choosing a career that is never abandoned. The second event is choosing a permanent job. Neal used the model to approximate and interpret observed career and job switches of young workers.

probability of occurring earlier in tenure than later), and (3) the probability of a subsequent quit is negatively correlated with the current wage rate. Jovanovic's insight was that each of these empirical regularities could be interpreted as reflecting the operation of a matching process with gradual learning about match quality.

A market has two sides that could be variously interpreted as consisting of firms and workers, or men and women, or owners and renters, or lakes and fishermen.[20] Following Jovanovic, we shall adopt the firm-worker interpretation here. An unmatched worker and a firm form a pair and jointly draw a random match parameter θ from a probability distribution with cumulative distribution function $\text{Prob}\{\theta \leq s\} = F(s)$. Here the match parameter reflects the marginal productivity of the worker in the match. In the first period, before the worker decides whether to work at this match or to wait and to draw a new match next period from the same distribution F, the worker and the firm both observe only $y = \theta + u$, where the random noise u is uncorrelated with θ. Thus, in the first period, the worker-firm pair receives only a noisy observation on θ and therefore both worker and firm form only an error-ridden impression of the quality of the match at first. On the basis of this noisy observation, the firm, which is imagined to operate competitively under constant returns to scale, offers to pay the worker the conditional expectation of θ, given $(\theta + u)$, for the first period, with the understanding that in subsequent periods it will pay the worker the expected value of θ, depending on whatever additional information both sides of the match receive.[21] Given this policy of the firm, the worker decides whether to accept the match and to work this period for $E[\theta|(\theta + u)]$ or to refuse the offer and draw a new match parameter θ' and noisy observation on it, $(\theta' + u')$, next period. If the worker decides to accept the offer in the first period, then in the second period both the firm and the worker are assumed to observe the true value of θ, meaning that both sides of the match learn about each other and about the quality of the match. In the second period the firm offers to pay the worker θ then and forever more. The worker next decides whether to

[20] We consider a simplified version of Jovanovic's model of matching. Prescott and Townsend, 1980, describe a discrete-time version of Jovanovic's model that we have further simplified here.

[21] Jovanovic assumed firms to be risk neutral and to maximize the expected present value of profits. They compete for workers by offering wage contracts. In a long-run equilibrium the payments practices of each firm would be well understood, and this fact would support the described implicit contract as a competitive equilibrium.

accept this offer or to quit, be unemployed this period, and draw a new match parameter and a noisy observation on it next period.

We can conveniently think of this process as having three stages. Stage 1 is the "predraw" stage, in which a previously unemployed worker has yet to draw the one match parameter and the noisy observation on it that he is entitled to draw after being unemployed the previous period. We let Q denote the expected present value of wages, before drawing, of a worker who was unemployed last period and who behaves optimally. The second stage of the process occurs after the worker has drawn a match parameter θ, has received the noisy observation of $(\theta + u)$ on it, and has received the firm's wage offer of $E[\theta|(\theta + u)]$ for this period. At this stage, the worker decides whether to accept this wage for this period and the prospect of receiving θ in all subsequent periods. The third stage occurs in the next period, when the worker and firm discover the true value of θ and the worker must decide whether to work at θ this period and in all subsequent periods that he remains at this job (match).

We now add some more specific assumptions about the probability distribution of θ and u. We assume that θ and u are independently distributed random variables. Both are normally distributed, θ being normal with mean μ and variance σ_0^2, and u being normal with mean 0 and variance σ_u^2. Thus, we write

$$\theta \sim N\left(\mu, \sigma_0^2\right), \qquad u \sim N\left(0, \sigma_u^2\right) \ . \tag{7.7.1}$$

In the first period, after drawing a θ, the worker and firm both observe the noise-ridden version of θ, $y = \theta + u$. Both worker and firm are interested in making inferences about θ, given the observation $(\theta + u)$. They are assumed to use Bayes' law and to calculate the posterior probability distribution of θ, that is, the probability distribution of θ conditional on $(\theta + u)$. The probability distribution of θ, given $\theta + u = y$, is known to be normal, with mean m_0 and variance σ_1^2. Using formulas for conditional distributions from section 2.5 or the Kalman filtering formula from section 2.7 in chapter 2, we have[22]

$$\begin{aligned} m_0 &= E(\theta|y) = E(\theta) + \frac{\operatorname{cov}(\theta, y)}{\operatorname{var}(y)}[y - E(y)] \\ &= \mu + \frac{\sigma_0^2}{\sigma_0^2+\sigma_u^2}(y-\mu) \equiv \mu + K_0(y-\mu)\ , \qquad (7.7.2) \\ \sigma_1^2 &= E\left[(\theta - m_0)^2 |y\right] = \frac{\sigma_0^2}{\sigma_0^2+\sigma_u^2}\sigma_u^2 = K_0\sigma_u^2 \,. \end{aligned}$$

After drawing θ and observing $y = \theta + u$ the first period, the firm is assumed to offer the worker a wage of $m_0 = E[\theta|(\theta+u)]$ the first period and a promise to pay θ for the second period and thereafter. The worker can accept or reject the offer.

From equation (7.7.2) and the property that the random variable $y - \mu = \theta + u - \mu$ is normal, with mean zero and variance $(\sigma_0^2 + \sigma_u^2)$, it follows that m_0 is itself normally distributed, with mean μ and variance $\sigma_0^4/(\sigma_0^2+\sigma_u^2) = K_0\sigma_0^2$:

$$m_0 \sim N\left(\mu, K_0\sigma_0^2\right). \qquad (7.7.3)$$

Note that $K_0\sigma_0^2 < \sigma_0^2$, so that m_0 has the same mean but a smaller variance than θ.

7.7.1. Recursive formulation

The worker seeks to maximize the expected present value of wages. We now proceed to solve the worker's problem by working backward. At stage 3, the worker knows θ and is confronted by the firm with an offer to work this period and forever more at a wage of θ. We let $J(\theta)$ be the expected present value of wages of a worker at stage 3 who has a known match θ in hand and who behaves optimally. The worker who accepts the match this period receives θ this period and faces the same choice at the same θ next period. (The worker can quit next period, though it will turn out that the worker who does not quit this period never will.) Therefore, if the worker accepts the match, the value of match θ is given by $\theta + \beta J(\theta)$, where β is the discount factor. The worker who rejects the match must be unemployed this period and must draw a new match next

[22] In the special case in which random variables are jointly normally distributed, linear least-squares projections equal conditional expectations.

period. The expected present value of wages of a worker who was unemployed last period and who behaves optimally is Q. Therefore, the Bellman equation is $J(\theta) = \max\{\theta + \beta J(\theta), \beta Q\}$. This equation is graphed in Figure 7.7.1 and evidently has the solution

$$J(\theta) = \begin{cases} \theta + \beta J(\theta) = \frac{\theta}{1-\beta} & \text{for } \theta \geq \overline{\theta} \\ \beta Q & \text{for } \theta \leq \overline{\theta}. \end{cases} \tag{7.7.4}$$

The optimal policy is a reservation wage policy: accept offers $\theta \geq \overline{\theta}$, and reject offers $\theta \leq \overline{\theta}$, where θ satisfies

$$\frac{\overline{\theta}}{1-\beta} = \beta Q. \tag{7.7.5}$$

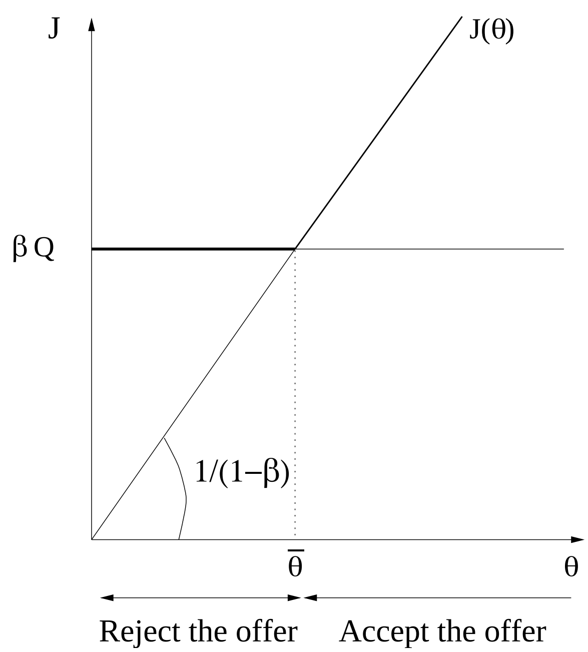

Figure 7.7.1: The function $J(\theta) = \max\{\theta + \beta J(\theta), \beta Q\}$. The reservation wage in stage 3, $\overline{\theta}$, satisfies $\overline{\theta}/(1-\beta) = \beta Q$.

We now turn to the worker's decision in stage 2, given the decision rule in stage 3. In stage 2, the worker is confronted with a current wage offer $m_0 =$

$E[\theta|(\theta+u)]$ and a conditional probability distribution function that we write as Prob$\{\theta \leq s|\theta+u\} = F(s|m_0, \sigma_1^2)$. (Because the distribution is normal, it can be characterized by the two parameters m_0, σ_1^2.) We let $V(m_0)$ be the expected present value of wages of a worker at the second stage who has offer m_0 in hand and who behaves optimally. The worker who rejects the offer is unemployed this period and draws a new match parameter next period. The expected present value of this option is βQ. The worker who accepts the offer receives a wage of m_0 this period and a probability distribution of wages of $F(\theta'|m_0, \sigma_1^2)$ for next period. The expected present value of this option is $m_0+\beta\int J(\theta')dF(\theta'|m_0,\sigma_1^2)$. The Bellman equation for the second stage therefore becomes

$$V\left(m_0\right) = \max\left\{m_0 + \beta\int J\left(\theta'\right)dF\left(\theta'|m_0,\sigma_1^2\right), \beta Q\right\}. \tag{7.7.6}$$

Note that both m_0 and $\beta\int J(\theta')dF(\theta'|m_0,\sigma_1^2)$ are increasing in m_0, whereas βQ is a constant. For this reason a reservation wage policy will be an optimal one. The functional equation evidently has the solution

$$V\left(m_0\right) = \begin{cases} m_0 + \beta\int J\left(\theta'\right)dF\left(\theta'|m_0,\sigma_1^2\right) & \text{for } m_0 \geq \overline{m}_0 \\ \beta Q & \text{for } m_0 \leq \overline{m}_0. \end{cases} \tag{7.7.7}$$

If we use equation (7.7.7), an implicit equation for the reservation wage $\overline{m}_0$ is then

$$V\left(\overline{m}_0\right) = \overline{m}_0 + \beta\int J\left(\theta'\right)dF\left(\theta'|\overline{m}_0,\sigma_1^2\right) = \beta Q. \tag{7.7.8}$$

Using equations (7.7.8) and (7.7.4), we shall show that $\overline{m}_0 < \overline{\theta}$, so that the worker becomes choosier over time with the firm. This force makes wages rise with tenure.

Using equations (7.7.4) and (7.7.5) repeatedly in equation (7.7.8), we obtain

$$\begin{aligned}
&\overline{m}_0 + \beta\frac{\overline{\theta}}{1-\beta}\int_{-\infty}^{\overline{\theta}} dF\left(\theta'|\overline{m}_0,\sigma_1^2\right) + \frac{\beta}{1-\beta}\int_{\overline{\theta}}^{\infty}\theta' dF\left(\theta'|\overline{m}_0,\sigma_1^2\right) \\
&\quad = \frac{\overline{\theta}}{1-\beta} = \frac{\overline{\theta}}{1-\beta}\int_{-\infty}^{\overline{\theta}} dF\left(\theta'|\overline{m}_0,\sigma_1^2\right) \\
&\quad + \frac{\overline{\theta}}{1-\beta}\int_{\overline{\theta}}^{\infty} dF\left(\theta'|\overline{m}_0,\sigma_1^2\right).
\end{aligned}$$

Rearranging this equation, we get

$$\overline{\theta}\int_{-\infty}^{\overline{\theta}} dF\left(\theta'|\overline{m}_0,\sigma_1^2\right)-\overline{m}_0=\frac{1}{1-\beta}\int_{\overline{\theta}}^{\infty}\left(\beta\theta'-\overline{\theta}\right)dF\left(\theta'|\overline{m}_0,\sigma_1^2\right). \tag{7.7.9}$$

Now note the identity

$$\overline{\theta}=\int_{-\infty}^{\overline{\theta}}\overline{\theta}dF\left(\theta'|\overline{m}_0,\sigma_1^2\right)+\left(\frac{1}{1-\beta}-\frac{\beta}{1-\beta}\right)\int_{\overline{\theta}}^{\infty}\overline{\theta}dF\left(\theta'|\overline{m}_0,\sigma_1^2\right). \tag{7.7.10}$$

Adding equation (7.7.10) to (7.7.9) gives

$$\overline{\theta}-\overline{m}_0=\frac{\beta}{1-\beta}\int_{\overline{\theta}}^{\infty}\left(\theta'-\overline{\theta}\right)dF\left(\theta'|\overline{m}_0,\sigma_1^2\right). \tag{7.7.11}$$

The right side of equation (7.7.11) is positive. The left side is therefore also positive, so that we have established that

$$\overline{\theta}>\overline{m}_0. \tag{7.7.12}$$

Equation (7.7.11) resembles equation (7.2.3) and has a related interpretation. Given $\overline{\theta}$ and $\overline{m}_0$, the right side is the expected benefit of a match $\overline{m}_0$, namely, the expected present value of the match in the event that the match parameter eventually turns out to exceed the reservation match $\overline{\theta}$ so that the match endures. The left side is the one-period cost of temporarily staying in a match paying less than the eventual reservation match value $\overline{\theta}$: having remained unemployed for a period in order to have the privilege of drawing the match parameter θ, the worker has made an investment to acquire this opportunity and must make a similar investment to acquire a new one. Having only the noisy observation of $(\theta+u)$ on θ, the worker is willing to stay in matches m_0 with $\overline{m}_0<m_0<\overline{\theta}$ because it is worthwhile to speculate that the match is really better than it seems now and will seem next period.

Now turning briefly to stage 1, we have defined Q as the predraw expected present value of wages of a worker who was unemployed last period and who is about to draw a match parameter and a noisy observation on it. Evidently, Q is given by

$$Q=\int V\left(m_0\right)dG\left(m_0|\mu,K_0\sigma_0^2\right) \tag{7.7.13}$$

where $G(m_0|\mu,K_0\sigma_0^2)$ is the normal distribution with mean μ and variance $K_0\sigma_0^2$, which, as we saw before, is the distribution of m_0.

Collecting some of the equations, we see that the worker's optimal policy is determined by

$$J(\theta) = \begin{cases} \theta + \beta J(\theta) = \frac{\theta}{1-\beta} & \text{for } \theta \geq \overline{\theta} \\ \beta Q & \text{for } \theta \leq \overline{\theta} \end{cases} \tag{7.7.14}$$

$$V(m_0) = \begin{cases} m_0 + \beta \int J(\theta')\, dF\left(\theta' | m_0, \sigma_1^2\right) & \text{for } m_0 \geq \overline{m}_0 \\ \beta Q & \text{for } m_0 \leq \overline{m}_0 \end{cases} \tag{7.7.15}$$

$$\overline{\theta} - \overline{m}_0 = \frac{\beta}{1-\beta} \int_{\overline{\theta}}^{\infty} \left(\theta' - \overline{\theta}\right) dF\left(\theta' | \overline{m}_0, \sigma_1^2\right) \tag{7.7.16}$$

$$Q = \int V(m_0)\, dG\left(m_0 | \mu, K_0 \sigma_0^2\right). \tag{7.7.17}$$

To analyze formally the existence and uniqueness of a solution to these equations, one would proceed as follows. Use equations (7.7.14), (7.7.15), and (7.7.16) to write a single functional equation in V,

$$\begin{aligned} V(m_0) = \max\Bigg\{ & m_0 + \beta \int \max\left[\frac{\theta}{1-\beta}\, , \right. \\ & \left. \beta \int V(m_1') dG\left(m_1' | \mu, K_0\sigma_0^2\right)\right] dF(\theta | m_0, \sigma_1^2)\, , \\ & \beta \int V(m_1') dG\left(m_1' | \mu, K_0\sigma_0^2\right) \Bigg\}. \end{aligned} \tag{7.7.18}$$

The expression on the right defines an operator T that maps continuous functions V into continuous functions TV. Th functional equation (7.7.18) can be expressed $V = TV$. The operator T can be directly verified to satisfy the following two properties: (1) it is monotone, that is, $v(m) \geq z(m)$ for all m implies $(Tv)(m) \geq (Tz)(m)$ for all m; (2) for all positive constants c, $T(v+c) \leq Tv + \beta c$. These are Blackwell's sufficient conditions for the functional equation $Tv = v$ to have a unique continuous solution. See Technical Appendix A on functional analysis at the end of this book.

7.7.2. Endogenous statistics

We now proceed to calculate probabilities and expectations of some interesting events and variables. The probability that a previously unemployed worker accepts an offer is given by

$$\text{Prob}\{m_0 \geq \overline{m}_0\} = \int_{\overline{m}_0}^{\infty} dG\left(m_0|\mu, K_0\sigma_0^2\right).$$

The probability that a previously unemployed worker accepts an offer and then quits the second period is given by

$$\text{Prob}\{\left(\theta \leq \overline{\theta}\right) \cap \left(m_0 \geq \overline{m}_0\right)\} = \int_{\overline{m}_0}^{\infty} \int_{-\infty}^{\overline{\theta}} dF\left(\theta|m_0, \sigma_1^2\right) dG\left(m_0|\mu, K_0\sigma_0^2\right).$$

The probability that a previously unemployed worker accepts an offer the first period and also elects not to quit the second period is given by

$$\text{Prob}\{\left(\theta \geq \overline{\theta}\right) \cap \left(m_0 \geq \overline{m}\right)\} = \int_{\overline{m}_0}^{\infty} \int_{\overline{\theta}}^{\infty} dF\left(\theta|m_0, \sigma_1^2\right) dG\left(m_0|\mu, K_0\sigma_0^2\right).$$

The mean wage of those employed the first period is given by

$$\overline{w}_1 = \frac{\int_{\overline{m}_0}^{\infty} m_0 \, dG\left(m_0|\mu, K_0\sigma_0^2\right)}{\int_{\overline{m}_0}^{\infty} dG\left(m_0|\mu, K_0\sigma_0^2\right)}, \tag{7.7.19}$$

whereas the mean wage of those workers who are in the second period of tenure is given by

$$\overline{w}_2 = \frac{\int_{\overline{m}_0}^{\infty} \int_{\overline{\theta}}^{\infty} \theta \, dF\left(\theta|m_0, \sigma_1^2\right) \, dG\left(m_0|\mu, K_0\sigma_0^2\right)}{\int_{\overline{m}_0}^{\infty} \int_{\overline{\theta}}^{\infty} dF\left(\theta|m_0, \sigma_1^2\right) \, dG\left(m_0|\mu, K_0\sigma_0^2\right)}. \tag{7.7.20}$$

We shall now prove that $\overline{w}_2 > \overline{w}_1$, so that wages rise with tenure. After substituting $m_0 \equiv \int \theta dF(\theta|m_0,\sigma_1^2)$ into equation (7.7.19),

$$\begin{aligned}
\overline{w}_1 &= \frac{\int_{\overline{m}_0}^{\infty}\int_{-\infty}^{\infty} \theta\, dF\left(\theta|m_0,\sigma_1^2\right)\, dG\left(m_0|\mu,K_0\sigma_0^2\right)}{\int_{\overline{m}_0}^{\infty} dG\left(m_0|\mu,K_0\sigma_0^2\right)} \\
&= \frac{1}{\int_{\overline{m}_0}^{\infty} dG\left(m_0|\mu,K_0\sigma_0^2\right)}\Bigg\{\int_{\overline{m}_0}^{\infty}\int_{-\infty}^{\overline{\theta}} \theta\, dF\left(\theta|m_0,\sigma_1^2\right)\, dG\left(m_0|\mu,K_0\sigma_0^2\right) \\
&\qquad + \overline{w}_2\int_{\overline{m}_0}^{\infty}\int_{\overline{\theta}}^{\infty} dF\left(\theta|m_0,\sigma_1^2\right)\, dG\left(m_0|\mu,K_0\sigma_0^2\right)\Bigg\} \\
&< \frac{\int_{\overline{m}_0}^{\infty}\left\{\overline{\theta}\, F\left(\overline{\theta}|m_0,\sigma_1^2\right) + \overline{w}_2\left[1-F\left(\overline{\theta}|m_0,\sigma_1^2\right)\right]\right\} dG\left(m_0|\mu,K_0\sigma_0^2\right)}{\int_{\overline{m}_0}^{\infty} dG\left(m_0|\mu,K_0\sigma_0^2\right)} \\
&< \overline{w}_2.
\end{aligned}$$

It is quite intuitive that the mean wage of those workers who are in the second period of tenure must exceed the mean wage of all employed in the first period. The former group is a subset of the latter group where workers with low productivities, $\theta < \overline{\theta}$, have left. Since the mean wages are equal to the true average productivity in each group, it follows that $\overline{w}_2 > \overline{w}_1$.

The model thus implies that "wages rise with tenure," both in the sense that mean wages rise with tenure and in the sense that $\overline{\theta} > \overline{m}_0$, which asserts that the lower bound on second-period wages exceeds the lower bound on first-period wages. That wages rise with tenure was observation 1 that Jovanovic sought to explain.

Jovanovic's model also explains observation 2, that quits are negatively correlated with tenure. The model implies that quits occur between the first and second periods of tenure. Having decided to stay for two periods, the worker never quits.

The model also accounts for observation 3, namely, that the probability of a subsequent quit is negatively correlated with the current wage rate. The probability of a subsequent quit is given by

$$\text{Prob}\{\theta' < \overline{\theta}|m_0\} = F\left(\overline{\theta}|m_0,\sigma_1^2\right),$$

which is evidently negatively correlated with m_0, the first-period wage. Thus, the model explains each observation that Jovanovic sought to interpret. In the version of the model that we have studied, a worker eventually becomes permanently matched with probability 1. If we were studying a population of such workers of fixed size, all workers would eventually be absorbed into the state of being permanently matched. To provide a mechanism for replenishing the stock of unmatched workers, one could combine Jovanovic's model with the "firing" model in section 7.2.5. By letting matches θ "go bad" with probability λ each period, one could presumably modify Jovanovic's model to get the implication that, with a fixed population of workers, a fraction would remain unmatched each period because of the dissolution of previously acceptable matches.

7.8. A longer horizon version of Jovanovic's model

Here we consider a $T+1$ period version of Jovanovic's model, in which learning about the quality of the match continues for T periods before the quality of the match is revealed by "nature." (Jovanovic assumed that $T=\infty$.) We use the recursive projection technique (the Kalman filter) of chapter 2 to handle the firm's and worker's sequential learning. The prediction of the true match quality can then easily be updated with each additional noisy observation.

A firm-worker pair jointly draws a match parameter θ at the start of the match, which we call the beginning of period 0. The value θ is revealed to the pair only at the beginning of the $(T+1)$th period of the match. After θ is drawn but before the match is consummated, the firm-worker pair observes $y_0 = \theta + u_0$, where u_0 is random noise. At the beginning of each period of the match, the worker-firm pair draws another noisy observation $y_t = \theta + u_t$ on the match parameter θ. The worker then decides whether or not to continue the match for the additional period. Let $y^t = \{y_0, \ldots, y_t\}$ be the firm's and worker's information set at time t. We assume that θ and u_t are independently distributed random variables with $\theta \sim \mathcal{N}(\mu, \Sigma_0)$ and $u_t \sim \mathcal{N}(0, \sigma_u^2)$. For $t \geq 0$ define $m_t = E[\theta | y^t]$ and $m_{-1} = \mu$. The conditional means m_t and variances $E(\theta - m_t)^2 = \Sigma_{t+1}$ can be computed with the Kalman filter via the formulas from chapter 2:

$$m_t = (1 - K_t)\, m_{t-1} + K_t y_t \tag{7.8.1a}$$

$$K_t = \frac{\Sigma_t}{\Sigma_t + R} \tag{7.8.1b}$$

$$\Sigma_{t+1} = \frac{\Sigma_t R}{\Sigma_t + R}, \tag{7.8.1c}$$

where $R = \sigma_u^2$ and Σ_0 is the unconditional variance of θ. The recursions are to be initiated from $m_{-1} = \mu$, and given Σ_0.

Using the formulas from chapter 2, we have that conditional on y^t, $m_{t+1} \sim \mathcal{N}(m_t, K_{t+1}\Sigma_{t+1})$ and $\theta \sim \mathcal{N}(m_t, \Sigma_{t+1})$ where Σ_0 is the unconditional variance of θ.

7.8.1. The Bellman equations

For $t \geq 0$, let $v_t(m_t)$ be the value of the worker's problem at the beginning of period t for a worker who optimally estimates that the match value is m_t after having observed y^t. At the start of period $T+1$, we suppose that the value of the match is revealed without error. Thus, at time T, $\theta \sim \mathcal{N}(m_T, \Sigma_{T+1})$. The firm-worker pair estimates θ by m_t for $t = 0, \ldots, T$, and by θ for $t \geq T+1$. Then the following functional equations characterize the solution of the problem:

$$v_{T+1}(\theta) = \max\left\{\frac{\theta}{1-\beta}, \beta Q\right\}, \tag{7.8.2}$$

$$v_T(m) = \max\left\{m + \beta \int v_{T+1}(\theta)\, dF(\theta \mid m, \Sigma_{T+1}), \beta Q\right\}, \tag{7.8.3}$$

$$v_t(m) = \max\left\{m + \beta \int v_{t+1}(m')\, dF(m' | m, K_{t+1}\Sigma_{t+1}), \beta Q\right\}, \quad t = 0, \ldots, T-1, \tag{7.8.4}$$

$$Q = \int v_0(m)\, dF(m | \mu, K_0\Sigma_0), \tag{7.8.5}$$

with K_t and Σ_t from the Kalman filter. Starting from v_{T+1} and reasoning backward, it is evident that the worker's optimal policy is to set reservation wages $\overline{m}_t, t = 0, \ldots, T$ that satisfy

$$\begin{aligned}
&\overline{m}_{T+1} = \overline{\theta} = \beta(1-\beta) Q, \\
&\overline{m}_T + \beta \int v_{T+1}(\theta)\, dF(\theta | \overline{m}_T, \Sigma_{T+1}) = \beta Q, \\
&\overline{m}_t + \beta \int v_{t+1}(m')\, dF(m' \mid \overline{m}_t, K_{t+1}\Sigma_{t+1}) = \beta Q, \qquad t = 1, \ldots, T-1.
\end{aligned} \tag{7.8.6}$$

To compute a solution to the worker's problem, we can define a mapping from Q into itself, with the property that a fixed point of the mapping is the optimal value of Q. Here is an algorithm:

a. Guess a value of Q, say Q^i with $i = 1$.

b. Given Q^i, compute sequentially the value functions in equations (7.8.2) through (7.8.4). Let the solutions be denoted $v^i_{T+1}(\theta)$ and $v^i_t(m)$ for $t = 0, \ldots, T$.

c. Given $v^i_1(m)$, evaluate equation (7.8.5) and call the solution $\tilde{Q}^i$.

d. For a fixed "relaxation parameter" $g \in (0,1)$, compute a new guess of Q from

$$Q^{i+1} = gQ^i + (1-g)\,\tilde{Q}^i .$$

e. Iterate on this scheme to convergence.

We now turn to the case where the true θ is never revealed by nature, that is, $T = \infty$. Note that $(\Sigma_{t+1})^{-1} = (\sigma_u^2)^{-1} + (\Sigma_t)^{-1}$, so $\Sigma_{t+1} < \Sigma_t$ and $\Sigma_{t+1} \to 0$ as $t \to \infty$. In other words, the accuracy of the prediction of θ becomes arbitrarily good as the information set y^t becomes large. Consequently, the firm and worker eventually learn the true θ, and the value function "at infinity" becomes

$$v_\infty(\theta) = \max\left\{\frac{\theta}{1-\beta}, \beta Q\right\},$$

and the Bellman equation for any finite tenure t is given by equation (7.8.4), and Q in equation (7.8.5) is the value of an unemployed worker. The optimal policy is a reservation wage $\overline{m}_t$, one for each tenure t. In fact, in the absence of a final date $T+1$ when θ is revealed by nature, the solution is actually a time-invariant policy function $\overline{m}(\sigma_t^2)$ with an acceptance and a rejection region in the space of (m, σ^2).

To compute a numerical solution when $T = \infty$, we would still have to rely on the procedure that we have outlined based on the assumption of some finite date when the true θ is revealed, say in period $\hat{T}+1$. The idea is to choose a sufficiently large $\hat{T}$ so that the conditional variance of θ at time $\hat{T}$, $\sigma^2_{\hat{T}}$, is close to zero. We then examine the approximation that $\sigma^2_{\hat{T}+1}$ is equal to zero. That is, equations (7.8.2) and (7.8.3) are used to truncate an otherwise infinite series of value functions.

7.9. Concluding remarks

The situations analyzed in this chapter are ones in which a currently unemployed worker rationally chooses to refuse an offer to work, preferring to remain unemployed today in exchange for better prospects tomorrow. The worker is voluntarily unemployed in one sense, having chosen to reject the current draw from the distribution of offers. In this model, the activity of unemployment is an investment incurred to improve the situation faced in the future. A theory in which unemployment is voluntary permits an analysis of the forces impinging on the choice to remain unemployed. Thus we can study the response of the worker's decision rule to changes in the distribution of offers, the rate of unemployment compensation, the number of offers per period, and so on.

Chapter 24 studies the optimal design of unemployment compensation. That issue is a trivial one in the present chapter with risk-neutral agents and no externalities. Here the government should avoid any policy that affects the workers' decision rules since it would harm efficiency, and the first-best way of pursuing distributional goals is through lump-sum transfers. In contrast, chapter 24 assumes risk-averse agents and incomplete insurance markets, which together with information asymmetries, make for an intricate contract design problem in the provision of unemployment insurance.

Chapter 30 presents various equilibrium models of search and matching. We study workers searching for jobs in an island model, workers and firms forming matches in a model with a "matching function," and how a medium of exchange can overcome the problem of "double coincidence of wants" in a search model of money.

A. Nonnegative random variables

This appendix describes elementary properties of probability distributions that are used extensively in search theory.

7.A.1. Nonnegative random variables

We begin with some properties of nonnegative random variables that possess finite first moments. Consider a random variable p with a cumulative probability distribution function $F(P)$ defined by $\text{Prob}\{p \leq P\} = F(P)$. We assume that $F(0) = 0$ so that p is nonnegative. We assume that F, a nondecreasing function, is continuous from the right. We also assume that there is an upper bound $B < \infty$ such that $F(B) = 1$, so that p is bounded with probability 1.

The mean of p, Ep, is defined by

$$Ep = \int_0^B p \; dF(p) . \tag{7.A.1}$$

Let $u = 1 - F(p)$ and $v = p$ and use the integration-by-parts formula $\int_a^b u \; dv = uv\Big|_a^b - \int_a^b v \; du$, to verify that

$$\int_0^B [1 - F(p)] \, dp = \int_0^B p \; dF(p) .$$

Thus, we have the following formula for the mean of a nonnegative random variable:[23]

$$Ep = \int_0^B [1 - F(p)] \, dp = B - \int_0^B F(p) \, dp. \tag{7.A.2}$$

Now consider two independent random variables p_1 and p_2 drawn from the same distribution F. Consider the event $\{(p_1 < p) \cap (p_2 < p)\}$, which by the independence assumption has probability $F(p)^2$. The event $\{(p_1 < p) \cap (p_2 < p)\}$

[23] To compute the variance $E(p - E(p))^2 = E(p^2) - (E(p))^2$, first compute Ep^2. By integrating by parts, we can verify that

$$Ep^2 = \int_0^B p^2 dF(p) = 2 \int_0^B p \, (1 - F(p)) \, dp.$$

is equivalent to the event $\{\max(p_1, p_2) < p\}$, where "max" denotes the maximum. Therefore, if we use formula $(7.A.2)$, the random variable $\max(p_1, p_2)$ has mean

$$E \max(p_1, p_2) = B - \int_0^B F(p)^2 \, dp. \tag{7.A.3}$$

Similarly, if $p_1, p_2, \ldots, p_n$ are n independent random variables each drawn from F, then $\mathrm{Prob}\{\max(p_1, p_2, \ldots, p_n) < p\} = F(p)^n$ and

$$M_n \equiv E \max(p_1, p_2, \ldots, p_n) = B - \int_0^B F(p)^n \, dp, \tag{7.A.4}$$

where M_n is defined as the expected value of the maximum of $p_1, \ldots, p_n$.

7.A.2. Mean-preserving spreads

Rothschild and Stiglitz introduced the idea of a mean-preserving spread as a convenient way to characterize the riskiness of two distributions with the same mean. Consider a class of non-negative random variables with identical means. Index the probability distributions for this class of random variables by a parameter r belonging to some set R. For the rth distribution we denote $\mathrm{Prob}\{p \leq P\} = F(P, r)$ and assume that $F(P, r)$ is differentiable with respect to r for all $P \in [0, B]$. We assume that there is a single finite B such that $F(B, r) = 1$ for all r in R and that $F(0, r) = 0$ for all r in R, so that we are considering a class of distributions R for nonnegative, bounded random variables.

From equation $(7.A.2)$, we have

$$Ep = B - \int_0^B F(p, r) \, dp. \tag{7.A.5}$$

Therefore, two distributions with the same value of $\int_0^B F(\theta, r) d\theta$ have identical means. We write this as the identical means condition:

(i) $$\int_0^B [F(\theta, r_1) - F(\theta, r_2)] \, d\theta = 0.$$

Two distributions r_1, r_2 are said to satisfy the *single-crossing property* if there exists a $\hat{\theta}$ with $0 < \hat{\theta} < B$ such that

(ii) $$F(\theta, r_2) - F(\theta, r_1) \leq 0 \, (\geq 0) \qquad \text{when} \quad \theta \geq (\leq) \, \hat{\theta}.$$

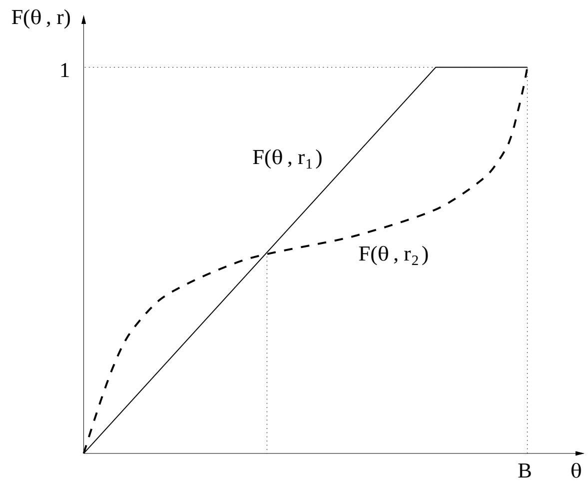

Figure 7.A.1: Two distributions, r_1 and r_2, that satisfy the single-crossing property.

Figure 7.A.1 illustrates the single-crossing property. If two distributions r_1 and r_2 satisfy properties (i) and (ii), we can regard distribution r_2 as having been obtained from r_1 by a process that shifts probability toward the tails of the distribution while keeping the mean constant.

Properties (i) and (ii) imply the following property:

$$\text{(iii)} \qquad \int_0^y \left[F\left(\theta, r_2\right) - F\left(\theta, r_1\right)\right] d\theta \geq 0, \qquad 0 \leq y \leq B\,.$$

Rothschild and Stiglitz regard properties (i) and (iii) as defining the concept of a "mean-preserving spread." In particular, a distribution indexed by r_2 is said to have been obtained from a distribution indexed by r_1 by a mean-preserving spread if the two distributions satisfy (i) and (iii).[24]

[24] Rothschild and Stiglitz (1970, 1971) use properties (i) and (iii) to characterize mean-preserving spreads rather than (i) and (ii) because (i) and (ii) fail to possess transitivity. That is, if $F(\theta, r_2)$ is obtained from $F(\theta, r_1)$ via a mean-preserving spread in the sense that the term has in (i) and (ii), and $F(\theta, r_3)$ is obtained from $F(\theta, r_2)$ via a mean-preserving spread in the sense of (i) and (ii), it does not follow that $F(\theta, r_3)$ satisfies the single-crossing property (ii) vis-à-vis distribution $F(\theta, r_1)$. A definition based on (i) and (iii), however, does provide a transitive ordering, which is a desirable feature for a definition designed to order distributions according to their riskiness.

For infinitesimal changes in r, Diamond and Stiglitz use the differential versions of properties (i) and (iii) to rank distributions with the same mean in order of riskiness. An increment in r is said to represent a mean-preserving increase in risk if

$$\text{(iv)} \qquad \int_0^B F_r\,(\theta, r)\,d\theta = 0$$

$$\text{(v)} \qquad \int_0^y F_r\,(\theta, r)\,d\theta \geq 0, \qquad 0 \leq y \leq B\ ,$$

where $F_r(\theta, r) = \partial F(\theta, r)/\partial r$.

B. More numerical dynamic programming

This appendix describes two more examples using the numerical methods of chapter 4.

7.B.1. Example 4: search

An unemployed worker wants to maximize $E_0 \sum_{t=0}^{\infty} \beta^t y_t$ where $y_t = w$ if the worker is employed at wage w, $y_t = 0$ if the worker is unemployed, and $\beta \in (0,1)$. Each period an unemployed worker draws a positive wage from a discrete-state Markov chain with transition matrix P. Thus, wage offers evolve according to a Markov process with transition probabilities given by

$$P\,(i,j) = \text{Prob}\,(w_{t+1} = \tilde{w}_j | w_t = \tilde{w}_i)\,.$$

Once he accepts an offer, the worker works forever at the accepted wage. There is no firing or quitting. Let v be an $(n \times 1)$ vector of values v_i representing the optimal value of the problem for a worker who has offer $w_i, i = 1, \ldots, n$ in hand and who behaves optimally. The Bellman equation is

$$v_i = \max_{\text{accept,reject}} \left\{ \frac{w_i}{1-\beta}, \beta \sum_{j=1}^{n} P_{ij} v_j \right\}$$

or

$$v = \max\{\tilde{w}/(1-\beta), \beta P v\}.$$

Here $\tilde{w}$ is an $(n \times 1)$ vector of possible wage values. This matrix equation can be solved using the numerical procedures described earlier. The optimal policy depends on the structure of the Markov chain P. Under restrictions on P making w positively serially correlated, the optimal policy has the following reservation wage form: there is a $\overline{w}$ such that the worker should accept an offer w if $w \geq \overline{w}$.

7.B.2. Example 5: a Jovanovic model

Here is a simplified version of the search model of Jovanovic (1979a). A newly unemployed worker draws a job offer from a distribution given by $\mu_i = \text{Prob}(w_1 = \tilde{w}_i)$, where w_1 is the first-period wage. Let μ be the $(n \times 1)$ vector with ith component μ_i. After an offer is drawn, subsequent wages associated with the job evolve according to a Markov chain with time-varying transition matrices

$$P_t(i,j) = \text{Prob}(w_{t+1} = \tilde{w}_j | w_t = \tilde{w}_i),$$

for $t = 1, \ldots, T$. We assume that for times $t > T$, the transition matrices $P_t = I$, so that after T a job's wage does not change anymore with the passage of time. We specify the P_t matrices to capture the idea that the worker-firm pair is learning more about the quality of the match with the passage of time. For example, we might set

$$P_t = \begin{bmatrix} 1-q^t & q^t & 0 & 0 & \ldots & 0 & 0 \\ q^t & 1-2q^t & q^t & 0 & \ldots & 0 & 0 \\ 0 & q^t & 1-2q^t & q^t & \ldots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \ldots & 1-2q^t & q^t \\ 0 & 0 & 0 & 0 & \ldots & q^t & 1-q^t \end{bmatrix},$$

where $q \in (0,1)$. In the following numerical examples, we use a slightly more general form of transition matrix in which (except at endpoints of the distribution),

$$\begin{aligned} \text{Prob}(w_{t+1} = \tilde{w}_{k\pm m} | w_t = \tilde{w}_k) = P_t(k, k \pm m) = q^t \\ P_t(k,k) = 1 - 2q^t. \end{aligned} \tag{7.B.1}$$

Here $m \geq 1$ is a parameter that indexes the spread of the distribution.

At the beginning of each period, a previously matched worker is exposed with probability $\lambda \in (0,1)$ to the event that the match dissolves. We then have a set of Bellman equations

$$v_t = \max\{\tilde{w} + \beta\,(1-\lambda)\,P_t v_{t+1} + \beta\lambda Q, \beta Q + \overline{c}\}, \tag{7.B.2a}$$

for $t = 1, \ldots, T,$ and

$$v_{T+1} = \max\{\tilde{w} + \beta\,(1-\lambda)\,v_{T+1} + \beta\lambda Q, \beta Q + \overline{c}\}, \tag{7.B.2b}$$

$$Q = \mu' v_1 \otimes \mathbf{1}$$
$$\overline{c} = c \otimes \mathbf{1}$$

where $\otimes$ is the Kronecker product, and $\mathbf{1}$ is an $(n \times 1)$ vector of ones. These equations can be solved by using calculations of the kind described previously. The optimal policy is to set a sequence of reservation wages $\{\overline{w}_j\}_{j=1}^{T}$.

Wage distributions

We can use recursions to compute probability distributions of wages at tenures $1, 2, \ldots, n$. Let the reservation wage for tenure j be $\overline{w}_j \equiv \tilde{w}_{\rho(j)}$, where $\rho(j)$ is the index associated with the cutoff wage. For $i \geq \rho(1)$, define

$$\delta_1\,(i) = \text{Prob}\,\{w_1 = \tilde{w}_i \mid w_1 \geq \overline{w}_1\} = \frac{\mu_i}{\sum_{h=\rho(1)}^{n} \mu_h}.$$

Then

$$\gamma_2\,(j) = \text{Prob}\,\{w_2 = \tilde{w}_j \mid w_1 \geq \overline{w}_1\} = \sum_{i=\rho(1)}^{n} P_1\,(i,j)\,\delta_1\,(i)\,.$$

For $i \geq \rho(2)$, define

$$\delta_2\,(i) = \text{Prob}\,\{w_2 = \tilde{w}_i \mid w_2 \geq \overline{w}_2 \cap w_1 \geq \overline{w}_1\}$$

or

$$\delta_2\,(i) = \frac{\gamma_2\,(i)}{\sum_{h=\rho(2)}^{n} \gamma_2\,(h)}.$$

Then

$$\gamma_3\,(j) = \text{Prob}\,\{w_3 = \tilde{w}_j \mid w_2 \geq \overline{w}_2 \cap w_1 \geq \overline{w}_1\} = \sum_{i=\rho(2)}^{n} P_2\,(i,j)\,\delta_2\,(i)\,.$$

Next, for $i \geq \rho(3)$, define $\delta_3(i) = \text{Prob}\{w_3 = \tilde{w}_i \mid (w_3 \geq \overline{w}_3) \cap (w_2 \geq \overline{w}_2) \cap (w_1 \geq \overline{w}_1)\}$. Then

$$\delta_3(i) = \frac{\gamma_3(i)}{\sum_{h=\rho(3)}^{n} \gamma_3(h)}.$$

Continuing in this way, we can define the wage distributions $\delta_1(i), \delta_2(i), \delta_3(i), \ldots$. The mean wage at tenure k is given by

$$\sum_{i \geq \rho(k)} \tilde{w}_i \delta_k(i).$$

Separation probabilities

The probability of rejecting a first period offer is $Q(1) = \sum_{h<\rho(1)} \mu_h$. The probability of separating at the beginning of period $j \geq 2$ is $Q(j) = \sum_{h<\rho(j)} \gamma_j(h)$.

Numerical examples

Figures 7.B.1, 7.B.2, and 7.B.3 report some numerical results for three versions of this model. For all versions, we set $\beta = .95, c = 0, q = .5$, and $T+1 = 21$. For all three examples, we used a wage grid with 60 equispaced points on the interval $[0, 10]$.

For the initial distribution μ we used the uniform distribution. We used a sequence of transition matrices of the form $(7.B.1)$, with a "gap" parameter of m. For the first example, we set $m = 6$ and $\lambda = 0$, while the second sets $m = 10$ and $\lambda = 0$ and third sets $m = 10$ and $\lambda = .1$.

Figure 7.B.1 shows the reservation wage falls as m increases from 6 to 10, and that it falls further when the probability of being fired λ rises from zero to .1. Figure 7.B.2 shows the same pattern for average wages. Figure 7.B.3 displays quit probabilities for the first two models. They fall with tenure, with shapes and heights that depend to some degree on m, λ.

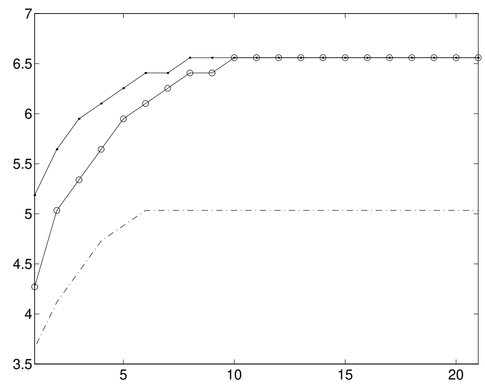

Figure 7.B.1: Reservation wages as a function of tenure for model with three different parameter settings $[m = 6, \lambda = 0]$ (the dots), $[m = 10, \lambda = 0]$ (the line with circles), and $[m = 10, \lambda = .1]$ (the dashed line).

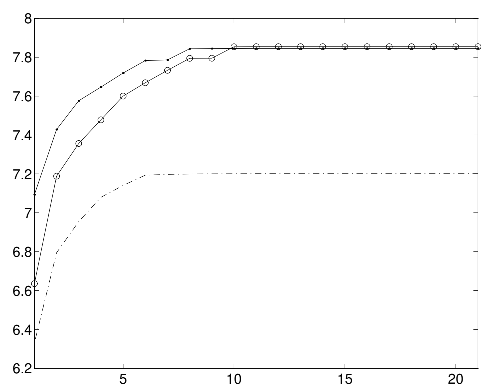

Figure 7.B.2: Mean wages as a function of tenure for model with three different parameter settings $[m = 6, \lambda = 0]$ (the dots), $[m = 10, \lambda = 0]$ (the line with circles), and $[m = 10, \lambda = .1]$ (the dashed line).

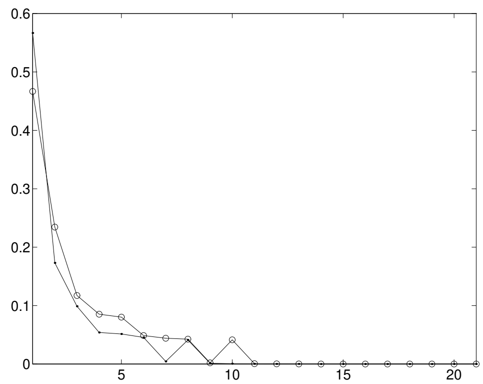

Figure 7.B.3: Quit probabilities as a function of tenure for Jovanovic model with $[m = 6, \lambda = 0]$ (line with dots) and $[m = 10, \lambda = .1]$ (the line with circles).

Exercises

Exercise 7.1 **Being unemployed with a chance of an offer**

An unemployed worker samples wage offers on the following terms: each period, with probability ϕ, $1 > \phi > 0$, she receives no offer (we may regard this as a wage offer of zero forever). With probability $(1-\phi)$ she receives an offer to work for w forever, where w is drawn from a cumulative distribution function $F(w)$. Assume that $F(0) = 0, F(B) = 1$ for some $B > 0$. Successive draws across periods are independently and identically distributed. The worker chooses a strategy to maximize

$$E\sum_{t=0}^{\infty} \beta^t y_t, \qquad \text{where} \quad 0 < \beta < 1,$$

$y_t = w$ if the worker is employed, and $y_t = c$ if the worker is unemployed. Here c is unemployment compensation, and w is the wage at which the worker is employed. Assume that, having once accepted a job offer at wage w, the worker stays in the job forever.

Let $v(w)$ be the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$ for an unemployed worker who has offer w in hand and who behaves optimally. Write the Bellman equation for the worker's problem.

Exercise 7.2 **Two offers per period**

Consider an unemployed worker who each period can draw *two* independently and identically distributed wage offers from the cumulative probability distribution function $F(w)$. The worker will work forever at the same wage after having once accepted an offer. In the event of unemployment during a period, the worker receives unemployment compensation c. The worker derives a decision rule to maximize $E\sum_{t=0}^{\infty}\beta^t y_t$, where $y_t = w$ or $y_t = c$, depending on whether she is employed or unemployed. Let $v(w)$ be the value of $E\sum_{t=0}^{\infty}\beta^t y_t$ for a currently unemployed worker who has best offer w in hand.

a. Formulate the Bellman equation for the worker's problem.

b. Prove that the worker's reservation wage is *higher* than it would be had the worker faced the same c and been drawing only *one* offer from the same distribution $F(w)$ each period.

Exercise 7.3 **A random number of offers per period**

An unemployed worker is confronted with a random number, n, of job offers each period. With probability π_n, the worker receives n offers in a given period, where $\pi_n \geq 0$ for $n \geq 1$, and $\sum_{n=1}^{N}\pi_n = 1$ for $N < +\infty$. Each offer is drawn independently from the same distribution $F(w)$. Assume that the number of offers n is independently distributed across time. The worker works forever at wage w after having accepted a job and receives unemployment compensation of c during each period of unemployment. He chooses a strategy to maximize $E\sum_{t=0}^{\infty}\beta^t y_t$ where $y_t = c$ if he is unemployed, $y_t = w$ if he is employed.

Let $v(w)$ be the value of the objective function of an unemployed worker who has best offer w in hand and who proceeds optimally. Formulate the Bellman equation for this worker.

Exercise 7.4 **Cyclical fluctuations in number of job offers**

Modify exercise *7.3* as follows: Let the number of job offers n follow a Markov process, with

$$\begin{aligned}&\text{Prob}\{\text{Number of offers next period} = m|\text{Number of offers this period} = n\}\\&\qquad\qquad = \pi_{mn}, \qquad m = 1,\ldots,N, \quad n = 1,\ldots,N\\&\sum_{m=1}^{N}\pi_{mn} = 1 \qquad \text{for} \quad n = 1,\ldots,N.\end{aligned}$$

Here $[\pi_{mn}]$ is a "stochastic matrix" generating a Markov chain. Keep all other features of the problem as in exercise *7.3*. The worker gets n offers per period, where n is now generated by a Markov chain so that the number of offers is possibly correlated over time.

a. Let $v(w,n)$ be the value of $E\sum_{t=0}^{\infty}\beta^t y_t$ for an unemployed worker who has received n offers this period, the best of which is w. Formulate the Bellman equation for the worker's problem.

b. Show that the optimal policy is to set a reservation wage $\overline{w}(n)$ that depends on the number of offers received this period.

Exercise 7.5 **Choosing the number of offers**

An unemployed worker must choose the number of offers n to solicit. At a cost of $k(n)$ the worker receives n offers this period. Here $k(n+1) > k(n)$ for $n \geq 1$. The number of offers n must be chosen in advance at the beginning of the period and cannot be revised during the period. The worker wants to maximize $E\sum_{t=0}^{\infty}\beta^t y_t$. Here y_t consists of w each period she is employed but not searching, $[w - k(n)]$ the first period she is employed but searches for n offers, and $[c - k(n)]$ each period she is unemployed but solicits and rejects n offers. The offers are each independently drawn from $F(w)$. The worker who accepts an offer works forever at wage w.

Let Q be the value of the problem for an unemployed worker who has not yet chosen the number of offers to solicit. Formulate the Bellman equation for this worker.

Exercise 7.6 **Mortensen externality**

Two parties to a match (say, worker and firm) jointly draw a match parameter θ from a c.d.f. $F(\theta)$. Once matched, they stay matched forever, each one deriving a benefit of θ per period from the match. Each unmatched pair of agents can influence the number of offers received in a period in the following way. The worker receives n offers per period, where $n = f(c_1 + c_2)$ and c_1 represents the resources the worker devotes to searching and c_2 represents the resources the typical firm devotes to searching. Symmetrically, the representative firm receives n offers per period where $n = f(c_1 + c_2)$. (We shall define the situation so that firms and workers have the same reservation θ so that there is never unrequited love.) Both c_1 and c_2 must be chosen at the beginning of the period, prior to searching during the period. Firms and workers have the same preferences, given by the expected present value of the match parameter θ, net of search costs. The discount factor β is the same for worker and firm.

a. Consider a Nash equilibrium in which party i chooses c_i, taking c_j, $j \neq i$, as given. Let Q_i be the value for an unmatched agent of type i before the level of c_i has been chosen. Formulate the Bellman equation for agents of types 1 and 2.

b. Consider the social planning problem of choosing c_1 and c_2 sequentially so as to maximize the criterion of λ times the utility of agent 1 plus $(1-\lambda)$ times the utility of agent 2, $0 < \lambda < 1$. Let $Q(\lambda)$ be the value for this problem for two unmatched agents before c_1 and c_2 have been chosen. Formulate the Bellman equation for this problem.

c. Comparing the results in a and b, argue that, in the Nash equilibrium, the optimal amount of resources has not been devoted to search.

Exercise 7.7 **Variable labor supply**

An unemployed worker receives each period a wage offer w drawn from the distribution $F(w)$. The worker has to choose whether to accept the job—and therefore to work forever—or to search for another offer and collect c in unemployment compensation. The worker who decides to accept the job must choose the number of hours to work in each period. The worker chooses a strategy to maximize

$$E \sum_{t=0}^{\infty} \beta^t u\left(y_t, l_t\right), \qquad \text{where} \quad 0 < \beta < 1,$$

and $y_t = c$ if the worker is unemployed, and $y_t = w(1-l_t)$ if the worker is employed and works $(1-l_t)$ hours; l_t is leisure with $0 \leq l_t \leq 1$.

Analyze the worker's problem. Argue that the optimal strategy has the reservation wage property. Show that the number of hours worked is the same in every period.

Exercise 7.8 **Wage growth rate and the reservation wage**

An unemployed worker receives each period an offer to work for wage w_t forever, where $w_t = w$ in the first period and $w_t = \phi^t w$ after t periods on the job. Assume $\phi > 1$, that is, wages increase with tenure. The initial wage offer is drawn from a distribution $F(w)$ that is constant over time (entry-level wages are stationary); successive drawings across periods are independently and identically distributed.

The worker's objective function is to maximize

$$E\sum_{t=0}^{\infty}\beta^t y_t, \qquad \text{where} \quad 0<\beta<1,$$

and $y_t = w_t$ if the worker is employed and $y_t = c$ if the worker is unemployed, where c is unemployment compensation. Let $v(w)$ be the optimal value of the objective function for an unemployed worker who has offer w in hand. Write the Bellman equation for this problem. Argue that, if two economies differ only in the growth rate of wages of employed workers, say $\phi_1 > \phi_2$, the economy with the higher growth rate has the smaller reservation wage. *Note:* Assume that $\phi_i\beta < 1$, $i = 1, 2$.

Exercise 7.9 **Search with a finite horizon**

Consider a worker who lives two periods. In each period the worker, if unemployed, receives an offer of lifetime work at wage w, where w is drawn from a distribution F. Wage offers are identically and independently distributed over time. The worker's objective is to maximize $E\{y_1 + \beta y_2\}$, where $y_t = w$ if the worker is employed and is equal to c—unemployment compensation—if the worker is not employed.

Analyze the worker's optimal decision rule. In particular, establish that the optimal strategy is to choose a reservation wage in each period and to accept any offer with a wage at least as high as the reservation wage and to reject offers below that level. Show that the reservation wage decreases over time.

Exercise 7.10 **Finite horizon and mean-preserving spread**

Consider a worker who draws every period a job offer to work forever at wage w. Successive offers are independently and identically distributed drawings from a distribution $F_i(w)$, $i = 1, 2$. Assume that F_1 has been obtained from F_2 by a mean-preserving spread. The worker's objective is to maximize

$$E\sum_{t=0}^{T}\beta^t y_t, \qquad 0<\beta<1,$$

where $y_t = w$ if the worker has accepted employment at wage w and is zero otherwise. Assume that both distributions, F_1 and F_2, share a common upper bound, B.

a. Show that the reservation wages of workers drawing from F_1 and F_2 coincide at $t = T$ and $t = T - 1$.

b. Argue that for $t \leq T - 2$ the reservation wage of the workers that sample wage offers from the distribution F_1 is higher than the reservation wage of the workers that sample from F_2.

c. Now introduce unemployment compensation: the worker who is unemployed collects c dollars. Prove that the result in part a no longer holds; that is, the reservation wage of the workers that sample from F_1 is higher than the one corresponding to workers that sample from F_2 for $t = T - 1$.

Exercise 7.11 **Pissarides' analysis of taxation and variable search intensity**

An unemployed worker receives each period a zero offer (or no offer) with probability $[1 - \pi(e)]$. With probability $\pi(e)$ the worker draws an offer w from the distribution F. Here e stands for effort—a measure of search intensity—and $\pi(e)$ is increasing in e. A worker who accepts a job offer can be fired with probability α, $0 < \alpha < 1$. The worker chooses a strategy, that is, whether to accept an offer or not and how much effort to put into search when unemployed, to maximize

$$E \sum_{t=0}^{\infty} \beta^t y_t, \qquad 0 < \beta < 1,$$

where $y_t = w$ if the worker is employed with wage w and $y_t = 1 - e + z$ if the worker spends e units of leisure searching and does not accept a job. Here z is unemployment compensation. For the worker who searched and accepted a job, $y_t = w - e - T(w)$; that is, in the first period the wage is net of search costs. Throughout, $T(w)$ is the amount paid in taxes when the worker is employed. We assume that $w - T(w)$ is increasing in w. Assume that $w - T(w) = 0$ for $w = 0$, that if $e = 0$, then $\pi(e) = 0$—that is, the worker gets no offers—and that $\pi'(e) > 0$, $\pi''(e) < 0$.

a. Analyze the worker's problem. Establish that the optimal strategy is to choose a reservation wage. Display the condition that describes the optimal choice of e, and show that the reservation wage is independent of e.

b. Assume that $T(w) = t(w - a)$ where $0 < t < 1$ and $a > 0$. Show that an increase in a decreases the reservation wage and increases the level of effort, increasing the probability of accepting employment.

c. Show under what conditions a change in t has the opposite effect.

Exercise 7.12 **Search and financial income**

An unemployed worker receives every period an offer to work forever at wage w, where w is drawn from the distribution $F(w)$. Offers are independently and identically distributed. Every agent has another source of income, which we denote ϵ_t, that may be regarded as financial income. In every period all agents get a realization of ϵ_t, which is independently and identically distributed over time, with distribution function $G(\epsilon)$. We also assume that w_t and ϵ_t are independent. The objective of a worker is to maximize

$$E\sum_{t=0}^{\infty}\beta^t y_t, \qquad 0<\beta<1,$$

where $y_t = w+\phi\epsilon_t$ if the worker has accepted a job that pays w, and $y_t = c+\epsilon_t$ if the worker remains unemployed. We assume that $0<\phi<1$ to reflect the fact that an employed worker has less time to collect financial income. Assume $1 > \text{Prob}\{w \geq c+(1-\phi)\epsilon\} > 0$.

Analyze the worker's problem. Write down the Bellman equation, and show that the reservation wage increases with the level of financial income.

Exercise 7.13 **Search and asset accumulation**

A previously unemployed worker receives an offer to work forever at wage w, but only if he chooses to do so, where w is drawn from the distribution $F(w)$. Previously employed workers receive no offers to work. But a previously employed worker is free to quit in any period, receive unemployment compensation that period, and so become a previously unemployed worker in the following period. Wage offers are identically and independently distributed over time. The worker maximizes

$$E\sum_{t=0}^{\infty}\beta^t\left(u\left(c_t\right)+v\left(l_t\right)\right), \qquad 0<\beta<1,$$

where c_t is consumption and l_t is leisure. Assume that $u(c)$ is strictly increasing, twice continuously differentiable, bounded, and strictly concave, while $v(l)$ is strictly increasing, twice continuously differentiable, and strictly concave; that $c_t \geq 0$; and that $l_t \in \{0,1\}$, so that the person can either work full time (here $l_t = 0$) or not at all (here $l_t = 1$). A gross return on assets a_t held between t

and $t+1$ is R_{t+1} and is i.i.d. with c.d.f. $H(R)$. The budget constraint is given by

$$a_{t+1} \leq R_{t+1}\,(a_t + w_t - c_t)$$

if the worker has a job that pays w_t. The random gross return R_{t+1} is observed at the beginning of period $t+1$ before the worker chooses n_{t+1}, c_{t+1}. If the worker is unemployed, the budget constraint is $a_{t+1} \leq R_{t+1}(a_t + z - c_t)$ and $l_t = 1$. Here z is unemployment compensation. It is assumed that a_t, the worker's asset position, cannot be negative. This is a no-borrowing assumption. Write a Bellman equation for this problem.

Exercise 7.14 **Temporary unemployment compensation**

Each period an unemployed worker draws one, and only one, offer to work forever at wage w. Wages are i.i.d. draws from the c.d.f. F, where $F(0) = 0$ and $F(B) = 1$. The worker seeks to maximize $E\sum_{t=0}^{\infty}\beta^t y_t$, where y_t is the sum of the worker's wage and unemployment compensation, if any. The worker is entitled to unemployment compensation in the amount $\gamma > 0$ only during the *first* period that she is unemployed. After one period on unemployment compensation, the worker receives none.

a. Write the Bellman equations for this problem. Prove that the worker's optimal policy is a time-varying reservation wage strategy.

b. Show how the worker's reservation wage varies with the duration of unemployment.

c. Show how the worker's "hazard of leaving unemployment" (i.e., the probability of accepting a job offer) varies with the duration of unemployment.

Now assume that the worker is also entitled to unemployment compensation if she quits a job. As before, the worker receives unemployment compensation in the amount of γ during the first period of an unemployment spell, and zero during the remaining part of an unemployment spell. (To qualify again for unemployment compensation, the worker must find a job and work for at least one period.)

The timing of events is as follows. At the very beginning of a period, a worker who was employed in the previous period must decide whether or not to quit. The decision is irreversible; that is, a quitter cannot return to an old job. If the worker quits, she draws a new wage offer as described previously, and if she accepts the offer she immediately starts earning that wage without suffering any period of unemployment.

d. Write the Bellman equations for this problem. *Hint*: At the very beginning of a period, let $v^e(w)$ denote the value of a worker who was employed in the previous period with wage w (before any wage draw in the current period). Let $v_1^u(w')$ be the value of an unemployed worker who has drawn wage offer w' and who is entitled to unemployment compensation, if she rejects the offer. Similarly, let $v_+^u(w')$ be the value of an unemployed worker who has drawn wage offer w' but who is not eligible for unemployment compensation.

e. Characterize the three reservation wages, $\overline{w}^e$, $\overline{w}_1^u$, and $\overline{w}_+^u$, associated with the value functions in part d. How are they related to γ? (*Hint*: Two of the reservation wages are straightforward to characterize, while the remaining one depends on the actual parameterization of the model.)

Exercise 7.15 **Seasons, I**

An unemployed worker seeks to maximize $E\sum_{t=0}^{\infty}\beta^t y_t$, where $\beta \in (0,1)$, y_t is her income at time t, and E is the mathematical expectation operator. The person's income consists of one of two parts: unemployment compensation of c that she receives each period she remains unemployed, or a fixed wage w that the worker receives if employed. Once employed, the worker is employed forever with no chance of being fired. Every odd period (i.e., $t = 1,3,5,\ldots$) the worker receives one offer to work forever at a wage drawn from the c.d.f. $F(W) = \text{Prob}(w \leq W)$. Assume that $F(0) = 0$ and $F(B) = 1$ for some $B > 0$. Successive draws from F are independent. Every even period (i.e., $t = 0,2,4,\ldots$), the unemployed worker receives two offers to work forever at a wage drawn from F. Each of the two offers is drawn independently from F.

a. Formulate the Bellman equations for the unemployed person's problem.

b. Describe the form of the worker's optimal policy.

Exercise 7.16 **Seasons, II**

Consider the following problem confronting an unemployed worker. The worker wants to maximize

$$E_0\sum_{t=0}^{\infty}\beta^t y_t, \quad \beta \in (0,1),$$

where $y_t = w_t$ in periods in which the worker is employed and $y_t = c$ in periods in which the worker is unemployed, where w_t is a wage rate and c is a constant level of unemployment compensation. At the start of each period, an unemployed worker receives one and only one offer to work at a wage w drawn from a c.d.f. $F(W)$, where $F(0) = 0, F(B) = 1$ for some $B > 0$. Successive draws from F are identically and independently distributed. There is no recall of past offers. Only unemployed workers receive wage offers. The wage is fixed as long as the worker remains in the job. The only way a worker can leave a job is if she is fired. At the *beginning* of each odd period $(t = 1, 3, \ldots)$, a previously employed worker faces the probability of $\pi \in (0, 1)$ of being fired. If a worker is fired, she immediately receives a new draw of an offer to work at wage w. At each even period $(t = 0, 2, \ldots)$, there is no chance of being fired.

a. Formulate a Bellman equation for the worker's problem.

b. Describe the form of the worker's optimal policy.

Exercise 7.17 **Gittins indexes for beginners**

At the end of each period,[25] a worker can switch between two jobs, A and B, to begin the following period at a wage that will be drawn at the beginning of next period from a wage distribution specific to job A or B, and to the worker's history of past wage draws from jobs of either type A or type B. The worker must decide to stay or leave a job at the end of a period after his wage for this period on his current job has been received, but before knowing what his wage would be next period in either job. The wage at either job is described by a job-specific n-state Markov chain. Each period the worker works at either job A or job B. At the end of the period, before observing next period's wage on either job, he chooses which job to go to next period. We use lowercase letters $(i, j = 1, \ldots, n)$ to denote states for job A, and uppercase letters $(I, J = 1, \ldots n)$ for job B. There is no option of being unemployed.

Let $w_a(i)$ be the wage on job A when state i occurs and $w_b(I)$ be the wage on job B when state I occurs. Let $A = [A_{ij}]$ be the matrix of one-step transition probabilities between the states on job A, and let $B = [B_{ij}]$ be the matrix for job B. If the worker leaves a job and later decides to return to it, he draws the wage for his first new period on the job from the conditional distribution determined by his last wage working at that job.

[25] See Gittins (1989) for more general versions of this problem.

The worker's objective is to maximize the expected discounted value of his lifetime earnings, $E_0 \sum_{t=0}^{\infty} \beta^t y_t$, where $\beta \in (0,1)$ is the discount factor, and where y_t is his wage from whichever job he is working at in period t.

a. Consider a worker who has worked at both jobs before. Suppose that $w_a(i)$ was the last wage the worker receives on job A and $w_b(I)$ the last wage on job B. Write the Bellman equation for the worker.

b. Suppose that the worker is just entering the labor force. The first time he works at job A, the probability distribution for his initial wage is $\pi_a = (\pi_{a1}, \ldots, \pi_{an})$. Similarly, the probability distribution for his initial wage on job B is $\pi_b = (\pi_{b1}, \ldots, \pi_{bn})$ Formulate the decision problem for a new worker, who must decide which job to take initially. *Hint*: Let $v_a(i)$ be the expected discounted present value of lifetime earnings for a worker who was last in state i on job A and has never worked on job B; define $v_b(I)$ symmetrically.

Exercise 7.18 **Jovanovic (1979b)**

An employed worker in the tth period of tenure on the current job receives a wage $w_t = x_t(1 - \phi_t - s_t)$ where x_t is job-specific human capital, $\phi_t \in (0,1)$ is the fraction of time that the worker spends investing in job-specific human capital, and $s_t \in (0,1)$ is the fraction of time that the worker spends searching for a new job offer. If the worker devotes s_t to searching at t, then with probability $\pi(s_t) \in (0,1)$ at the beginning of $t+1$ the worker receives a new job offer to begin working at new job-specific capital level μ' drawn from the c.d.f. $F(\cdot)$. That is, searching for a new job offer promises the prospect of instantaneously reinitializing job-specific human capital at μ'. Assume that $\pi'(s) > 0, \pi''(s) < 0$. While on a given job, job-specific human capital evolves according to

$$x_{t+1} = G(x_t, \phi_t) = g(x_t \phi_t) - \delta x_t,$$

where $g'(\cdot) > 0, g''(\cdot) < 0$, $\delta \in (0,1)$ is a depreciation rate, and $x_0 = \mu$ where t is tenure on the job, and μ is the value of the "match" parameter drawn at the start of the current job. The worker is risk neutral and seeks to maximize $E_0 \sum_{\tau=0}^{\infty} \beta^{\tau} y_{\tau}$, where y_τ is his wage in period τ.

a. Formulate the worker's Bellman equation.

b. Describe the worker's decision rule for deciding whether to accept an offer μ' at the beginning of next period.

c. Assume that $g(x\phi) = A(x\phi)^\alpha$ for $A > 0, \alpha \in (0,1)$. Assume that $\pi(s) = s^{.5}$. Assume that F is a discrete n-valued distribution with probabilities f_i; for example, let $f_i = n^{-1}$. Write a Matlab program to solve the Bellman equation. Compute the optimal policies for ϕ, s and display them.

Exercise 7.19 **Value function iteration and policy improvement algorithm**, donated by Pierre-Olivier Weill

The goal of this exercise is to study, in the context of a specific problem, two methods for solving dynamic programs: value function iteration and Howard's policy improvement. Consider McCall's model of intertemporal job search. An unemployed worker draws one offer from a c.d.f. F, with $F(0) = 0$ and $F(B) = 1$, $B < \infty$. If the worker rejects the offer, she receives unemployment compensation c and can draw a new wage offer next period. If she accepts the offer, she works forever at wage w. The objective of the worker is to maximize the expected discounted value of her earnings. Her discount factor is $0 < \beta < 1$.

a. Write the Bellman equation. Show that the optimal policy is of the reservation wage form. Write an equation for the reservation wage w^*.

b. Consider the value function iteration method. Show that at each iteration, the optimal policy is of the reservation wage form. Let w_n be the reservation wage at iteration n. Derive a recursion for w_n. Show that w_n converges to w^* at rate β.

c. Consider Howard's policy improvement algorithm. Show that at each iteration, the optimal policy is of the reservation wage form. Let w_n be the reservation wage at iteration n. Derive a recursion for w_n. Show that the rate of convergence of w_n towards w^* is locally quadratic. Specifically use a Taylor expansion to show that, for w_n close enough to w^*, there is a constant K such that $w_{n+1} - w^* \cong K(w_n - w^*)^2$.

Exercise 7.20

Different types of unemployed workers are identical, except that they sample from different wage distributions. Each period an unemployed worker of type α draws a single new offer to work forever at a wage w from a cumulative distribution function F_α that satisfies $F_\alpha(w) = 0$ for $w < 0$, $F_\alpha(0) = \alpha, F_\alpha(B) = 1$, where $B > 0$ and F_α is a right continuous function mapping $[0, B]$ into $[0, 1]$.

The c.d.f. of a type α worker is given by

$$F_\alpha(w) = \begin{cases} \alpha & \text{for } 0 \le w \le \alpha B \ ; \\ w/B & \text{for } \alpha B < w < B - \alpha B \ ; \\ 1-\alpha & \text{for } B - \alpha B \le w < B; \\ 1 & \text{for } w = B \end{cases}$$

where $\alpha \in [0, .5)$. An unemployed α worker seeks to maximize the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$, where $\beta \in (0,1)$ and $y_t = w$ if the worker is employed and $y_t = c$ if he or she is unemployed, where $0 < c < B$ is a constant level of unemployment compensation. By choosing a strategy for accepting or rejecting job offers, the worker affects the distribution with respect to which the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$ is calculated. The worker cannot recall past offers. If a previously unemployed worker accepts an offer to work at wage w this period, he must work forever at that wage (there is neither quitting nor firing nor searching for a new job while employed).

a. Formulate a Bellman equation for a type α worker. Prove that the worker's optimal strategy is to set a time-invariant reservation wage.

b. Consider two types of workers, $\alpha = 0$ and $\alpha = .3$. Can you tell which type of worker has a higher reservation wage?

c. Which type of worker would you expect to find a job more quickly?

Exercise 7.21 **Searching for the lowest price**

A buyer wants to purchase an item at the lowest price, net of total search costs. At a cost of $c > 0$ per draw, the buyer can draw an offer to buy the item at a price p that is drawn from the c.d.f. $F(P) = \text{Prob}(p \le P)$ where P is a non-decreasing, right-continuous function with $F(\underline{B}) = 0, F(\overline{B}) = 1$, where $0 < \underline{B} < \overline{B} < +\infty$. All search occurs within one period.

a. Find the buyer's optimal strategy.

b. Find an expression for the expected value of the purchase price net of all search costs.

Exercise 7.22 **Quits**

Each period an unemployed worker draws one offer to work at a nonnegative wage w, where w is governed by a c.d.f F that satisfies $F(0) = 0$ and $F(B) = 1$ for some $B > 0$. The worker seeks to maximize the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$

where $y_t = w$ if the worker is employed and c if the worker is unemployed. At the beginning of each period a worker employed at wage w the previous period is exposed to probability $\alpha \in (0,1)$ of having his job reclassified, which means that he will be given a new wage w' drawn from F. A reclassified worker has the option of working at wage w' until reclassified again, or quitting, receiving unemployment compensation of c this period, and drawing a new wage offer the next period.

a. Formulate the Bellman equation for an unemployed worker.

b. Describe the decision rule for a previously *unemployed* worker.

c. Describe the decision rule for quitting or staying for a previously *employed* worker.

d. Describe how to compute the probability that a previously employed worker will quit.

Exercise 7.23 **A career ladder**

Each period a previously unemployed worker draws one offer to work at a nonnegative wage w, where w is governed by a c.d.f F that satisfies $F(0) = 0$ and $F(B) = 1$ for some $B > 0$. The worker seeks to maximize the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$ where $\beta \in (0,1)$ and $y_t = w$ if the worker is employed and c if the worker is unemployed. At the beginning of each period a worker employed at wage w the previous period is exposed to a probability of $\alpha \in (0,1)$ of getting a promotion, which means that he will be given a new wage γw where $\gamma > 1$. This new wage will prevail until a next promotion.

a. Formulate a Bellman equation for a previously employed worker.

b. Formulate a Bellman equation for a previously unemployed worker.

c. Describe the decision rule for an unemployed worker.

d. Describe the decision rule for a previously employed worker.

Exercise 7.24 **Human capital**

A previously unemployed worker draws one offer to work at a wage wh, where h is his level of human capital and w is drawn from a c.d.f. F where $F(0) = 0, F(B) = 1$ for $B > 0$. The worker retains w, but not h, so long as he remains in his current job (or employment spell). The worker knows his current level of h before he draws w. Wage draws are independent over time.

When employed, the worker's human capital h evolves according to a discrete state Markov chain on the space $[\bar{h}_1, \bar{h}_2, \ldots, \bar{h}_n]$ with transition density H^e where $H^e(i,j) = \text{Prob}[h_{t+1} = \bar{h}_j | h_t = \bar{h}_i]$. When unemployed, the worker's human capital h evolves according to a discrete state Markov chain on the same space $[\bar{h}_1, \bar{h}_2, \ldots, \bar{h}_n]$ with transition density H^u where $H^u(i,j) = \text{Prob}[h_{t+1} = \bar{h}_j | h_t = \bar{h}_i]$. The two transition matrices H^e and H^u are such that an employed worker's human capital grows probabilistically (meaning that it is more likely that next period's human capital will be higher than this period's) and an unemployed worker's human capital decays probabilistically (meaning that it is more likely that next period's human capital will be lower than this period's). An unemployed worker receives unemployment compensation of c per period. The worker wants to maximize the expected value of $\sum_{t=0}^{\infty} \beta^t y_t$ where $y_t = w h_t$ when employed, and c when unemployed. At the beginning of each period, employed workers receive their new human capital realization from the Markov chain H^e. Then they are free to quit, meaning that they surrender their previous w, retain their newly realized level of human capital but immediately become unemployed, and can immediately draw a new w from F. They can accept that new draw immediately or else choose to be unemployed for at least one period while waiting for new opportunities to draw one w offer per period from F.

a. Obtain a Bellman equation or Bellman equations for the worker's problem.

b. Describe qualitatively the worker's optimal decision rule. Do you think employed workers might ever decide to quit?

c. Describe an algorithm to solve the Bellman equation or equations.

Exercise 7.25 **Markov wages**

Each period, a previously unemployed worker draws one offer to work forever at wage w. The worker wants to maximize $E \sum_{t=0}^{\infty} \beta^t y_t$, where $\beta \in (0,1)$ and $y_t = c > 0$ if the worker is unemployed, and $y_t = w$ if the worker is employed. Quitting is not allowed and once hired the worker cannot be fired. Successive draws of the wage are from a Markov chain with transition probabilities arranged in the $n \times n$ transition matrix P with (i,j) element $P_{ij} = \text{Prob}(w_{t+1} = \overline{w}_j | w_t = \overline{w}_i)$ where $\overline{w}_1 < \overline{w}_2 < \cdots < \overline{w}_n$.

a. Construct a Bellman equation for the worker.

b. Can you prove that the worker's optimal strategy is to set a reservation wage?

c. Assume that $\beta = .95$, $c = 1$, $[\,\overline{w}_1 \quad \overline{w}_2 \quad \cdots \quad \overline{w}_n\,] = [\,1 \quad 2 \quad 3 \quad 4 \quad 5\,]$ and

$$P = \begin{bmatrix} .8 & .2 & 0 & 0 & 0 \\ .18 & .8 & .02 & 0 & 0 \\ .25 & .25 & 0 & .25 & .25 \\ 0 & 0 & .02 & .8 & .18 \\ 0 & 0 & 0 & .2 & .8 \end{bmatrix}.$$

Please write a Matlab or R or C++ program to solve the Bellman equation. Show the optimal policy function and the value function.

d. Assume that all parameters are the same as in part **c** except for β, which now equals .99. Please find the optimal policy function and the optimal value function.

e. Please discuss whether, why, and how your answers to parts **c** and **d** differ.

Exercise 7.26 **Neal model's Markov implications**

This will be yet another exercise that illustrates the theme that finding the state is an art. Consider the version of the Neal (1999) career choice model that we analyzed in the text. In the text, a worker's state is the job, career pair ϵ, θ with which the worker enters the period. Knowing the structure of the outcome and the optimal decision rule, let's use figure 7.6.2 to partition the state space (ϵ, θ) into three sets that define the three new states $s_t \in \{1, 2, 3\}$ that we'll use to define the states in a Markov chain. We say that $s_t = 1$ if the worker wants a "new life", i.e., he wants to draw a new job, career pair next period. We say that $s_t = 2$ if the worker wants a new job, i.e., if he is content with his career θ but wants to draw a new job ϵ' next period. We say that $s_t = 3$ if the worker plans to remain in his current job, career pair next period. Let $P_{ij} = \text{Prob}(s_{t+1} = j | s_t = i)$. Assume that the initial probability distribution is $\pi_0 = \text{Prob}(s_0 = 1) = 1$.

a. Show that P has the following structure

$$P = \begin{bmatrix} 1 - P_{12} - P_{13} & P_{12} & P_{13} \\ 0 & 1 - P_{23} & P_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

and please tell how to compute the nontrivial elements of P from the information used to compute figure 7.6.2.

b. Describe the invariant distribution(s) of the Markov chain (π_0, P).

c. Is the Markov chain ergodic?

d. A labor economist has a panel of quarterly observations on a cohort of N 1995 high school graduates who did not go to college but instead went to work immediately (so Neal's model seems to apply). The observations record dates at which workers switched jobs and whether or not those were also associated with new occupations. Please describe how the economist might use those data to estimate the probabilities P_{ij} in Neal's model.

Exercise 7.27 **Neal model with unemployment**

Consider the following modification of the Neal (1999) model. A worker chooses career-job (θ, ϵ) pairs subject to the following conditions. If employed, the worker's earnings at time t equal $\theta_t + \epsilon_t$, where θ_t is a component specific to a *career* and ϵ_t is a component specific to a particular *job*. If unemployed, the worker receives unemployment compensation equal to c. The worker maximizes $E \sum_{t=0}^{\infty} \beta^t y_t$ where $y_t = (\theta_t + \epsilon_t)$ if the worker is employed and $y_t = c$ if the worker is unemployed. A *career* is a draw of θ from c.d.f. F; a *job* is a draw of ϵ from c.d.f. G. Successive draws are independent, and $G(0) = F(0) = 0$, $G(B_\epsilon) = F(B_\theta) = 1$. The worker can draw a new career only if he also draws a new job. However, the worker is free to retain his existing career (θ), and to draw a new job (ϵ') next period. The worker decides at the beginning of a period whether to stay in a career-job pair inherited from the past, stay in the inherited career but draw a new job *for next period*, or draw a new career-job pair (θ', ϵ') *for next period*. If the worker decides to draw either a new θ' or a new ϵ' for next period, he or she must become unemployed this period.

a. Let $v(\theta, \epsilon)$ be the optimal value of the problem at the beginning of a period for a worker currently having inherited career-job pair (θ, ϵ) and who is about to decide whether to decide whether to become unemployed in order to draw a new career and or job next period. Formulate a Bellman equation.

b. Characterize the worker's optimal policy.

Exercise 7.28 **A functional equation**

Please read the following functional equation:

$$v(w) = \max\Big\{w + \beta(1-\pi)v(w) + \beta\int v(w')\,dF(w'),\, c + \beta\int v(w')\,dF(w')\Big\},$$

where $c > 0$, $\beta \in (0,1)$, and F is a cumulative distribution function for a bounded nonnegative random variable. Please completely describe a decision problem that is associated with this functional equation.

Exercise 7.29 **Coupled functional equations**

Assume that $c > 0, \alpha \in (0,1), \gamma \in (0,1)$, and that F is a cumulative distribution function for a bounded nonnegative random variable. Assume that $h_H > h_L > 0$. Please read the following coupled functional equations:

$$v(w, h_L) = \max\Big\{wh_L + \beta\gamma v(w, h_H) + \beta(1-\gamma)v(w, h_L),\, c + \beta\int v(w', h_L)\,dF(w')\Big\}$$

$$v(w, h_H) = wh_H + \beta\alpha\int v(w', h_L)\,dF(w') + \beta(1-\alpha)v(w, h_H)$$

Please completely describe a decision problem that is associated with this pair of coupled functional equations.

Exercise 7.30 **Another functional equation**

Please read the following equation:

$$w(x) = \max\Big\{\frac{x}{1-\beta},\, d + \beta\int w(s)\,[\alpha f(s) + (1-\alpha)g(s)]\,ds\Big\}$$

where $f \geq 0, g \geq 0, \int f(x)dx = 1, \int g(x)dx = 1, \alpha \in (0,1)$. Please describe a decision problem associated with this equation. Please interpret each of the objects $w, x, \beta, d, \alpha, s, f, g$.

Exercise 7.31 **A reverse engineering problem**

Please describe a decision problem whose optimal value is given by a function $v(w)$ that satisfies the following equations:

$$v(w) = \max\{w + \beta(1-\alpha)v(w) + \beta\alpha Q,\, c + \beta Q\}$$

$$Q = \int v(w)\,dF(w).$$

In your answer, please define and interpret c, β, F, Q and $v(w)$.

Exercise 7.32 **Another reverse engineering problem**

Please describe a decision problem whose optimal value is given by a function $v(w)$ that satisfies the following equations:

$$\begin{aligned} v(w) &= \max\left\{w + \beta(1-\alpha)v(w) + \beta\alpha Q_f, c + \beta Q\right\} \\ Q &= \int v(w)\, dF^2(w) \\ Q_f &= \int v(w)\, dF(w). \end{aligned}$$

In your answer, please define and interpret c, β, F, Q, Q_f and $v(w)$.

Exercise 7.33 **Bayes Law again**

Let $f : [0, B] \to \mathbf{R}^+$ be a known probability density function for a nonnegative random variable and let $g : [0, B] \to \mathbf{R}^+$ be another known probability density function that puts positive probabilities on the same events as f. Let $p_{-1}(\alpha) : [0, 1] \to \mathbf{R}+$ be another known probability density function. For $t \geq 0$, let

$$p_t(\alpha) = \frac{[\alpha f(w_t) + (1-\alpha) g(w_t)]\, p_{t-1}(\alpha)}{\int [\tilde{\alpha} f(w_t) + (1-\tilde{\alpha}) g(w_t)]\, p_{t-1}(\tilde{\alpha})\, d\tilde{\alpha}}\ , \tag{1}$$

and for $t \geq -1$, let

$$h(w_{t+1}; p_t(\alpha)) = \int [\alpha f(w_{t+1}) + (1-\alpha) g(w_{t+1})]\, p_t(\alpha)\, d\alpha. \tag{2}$$

a. Please interpret $p_{-1}(\alpha)$ and $p_t(\alpha)$.

b. Please interpret formula (1)

c. Please interpret formula (2).

d. Please describe a decision problem that is associated with the following equation in the unknown function $v(w_t, p_t(\alpha))$:

$$v(w_t, p_t(\alpha)) = \max\left\{\frac{w_t}{1-\beta}, c + \beta \int v(w_{t+1}; p_{t+1}(\tilde{\alpha}))\, h(w_{t+1}; p_t(\alpha))\, dw_{t+1} d\alpha\right\}.$$

Exercise 7.34 **Quits again**

A worker who lives during periods $t = 0, 1, 2, \ldots, +\infty$ has an opportunity to work in period t at wage $w_t \geq 0$ and disutility of labor $\xi_t \geq 0$. Each period a previously unemployed worker has the opportunity to draw a new wage from a c.d.f. F that satisfies $F(0) = 0, F(B) = 1$ for $B > 0$. In period t, a worker employed last period at wage w_{t-1} has the opportunity not to draw again but instead to retain w_{t-1} and to set $w_t = w_{t-1}$. But if the worker quits, she cannot draw a new offer this period and in the future she cannot recall that offer or any other past offer. A worker draws one offer from F each period she starts a period after being unemployed the period before. An employed worker receives time t utility $w_t - \xi_t$, where each period ξ_t is a nonnegative random variable drawn afresh from c.d.f. G that satisfies $G(0) = 0, G(D) = 1$. Successive draws of ξ_t from G are independently and identically distributed. Successive new draws of w from F are independently and identically distributed. A worker receives unemployment benefits $b \geq 0$ each period she is unemployed. Where $\beta \in (0, 1)$ and time t utility is

$$u_t = \begin{cases} w_t - \xi_t & \text{if working} \\ \text{b} & \text{if not working,} \end{cases}$$

a worker wants to maximize the mathematical expectation of $\sum_{t=0}^{\infty} \beta^t u_t$ by sequentially deciding whether to retain last period's job being employed at wage w_{t-1} or quit and become unemployed and receive b for a period and then next period draw a new wage offer w and disutility of working ξ.

a. Please tell how the following functional equation generates a statistical model of quits and exits from unemployment:

$$\begin{aligned} v(w_t, \xi_t) &= \max_{\text{accept,reject}} \{w_t - \xi_t + \beta p(w_t), b + \beta Q\} \\ p(w_t) &= \int v(w_t, \xi') \, dG(\xi') \\ Q &= \int v(w, \xi) \, dF(w) \, dG(\xi) \, . \end{aligned} \tag{1}$$

Please carefully describe the objects $v(w, \xi), p(w),$ and Q.

b. Please describe a previously employed worker's optimal policy for quitting.

c. Please describe a previously unemployed worker's optimal policy for accepting a new job.

d. *Extra Credit* Assume that G and F are both uniform distributions. Write Python, Julia, or Matlab code to solve functional equation (1) when $b = B/4$, $D = B/2$, and $\beta = .95$. Describe the worker's optimal decisions quantitatively.